MURALPRESENTER: Multi-Agent Units of Responsibility for Authoring Long-Horizon Presentations

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Abstract

Automated presentation authoring is a long-horizon multimodal task requiring accurate slide content, effective layouts, and a coherent deck-level narrative. Deck-wide production concentrates execution history in one author, while per-slide production must coordinate cross-slide decisions that remain coupled. We introduce **MURALPRESENTER**, a skill-guided multi-agent framework for HTML presentation authoring and revision. The framework stores established facts, plans, and design decisions in shared persistent state. Its orchestrator assigns independent pages to Single Owners and pages requiring joint adjustment to a Group Owner, with responsibility extending through authoring, inspection, and revision. We train 9B and 27B variants on 9,000 curated execution segments and introduce **THREADEVAL** to assess intermediate execution and handoffs separately from final presentation quality. We further evaluate generation and editing on PresentBench, SlidesGen-Bench, and DECKBench. In a matched production-stage comparison on 35 tasks, per-slide and adaptive ownership improve Final scores over whole-deck ownership by 2.40 and 2.49 points, respectively, with greater recorded workload. The additional benefit of adaptive grouping over per-slide ownership remains uncertain. In a separate descriptive supervision study, Global Selective records 38.0% fewer input tokens and 30.2% fewer agent turns than Full on selected completed runs, while their Final-score difference remains uncertain. MURALPRESENTER-27B achieves 94.92% content accuracy and an Engagement score of 8.42 on SlidesGen-Bench, and completes all 294 five-turn DECKBench tasks. Inference code is available in the project repository.

1 Introduction

Presentation authoring transforms complex knowledge, source materials, and user requirements into a visual narrative that unfolds across slides. Its coherence depends on relationships beyond any single page: later comparisons must respect earlier numerical assumptions, conclusions must address opening questions with supporting evidence, and color encodings must retain their meaning throughout the deck. These relationships need to survive both initial production and subsequent revision, since individually readable slides can still omit requirements or contradict one another (Chen et al., 2026).

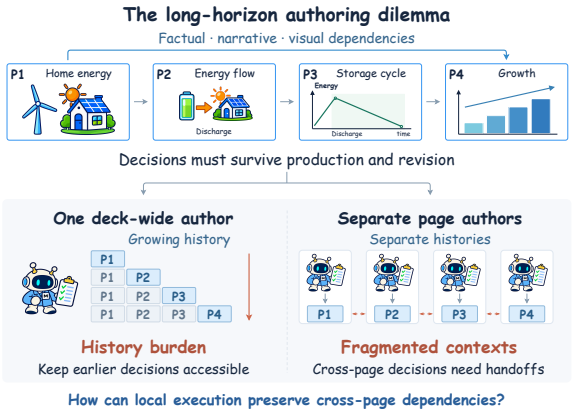


Figure 1: **The long-horizon authoring dilemma.** Cross-page decisions must survive production and revision. A deck-wide author accumulates execution history; separate page authors distribute local work but require handoffs to coordinate cross-page decisions.

Long-horizon agents accumulate plans, tool results, and revision history, yet a larger context does not guarantee reliable use of earlier information (Liu et al., 2024). External state and bounded continuation help manage this history (Sun et al., 2025; Ma et al., 2026). Multi-agent systems can distribute execution, but their benefits depend on task structure and coordination costs (Zhuge et al., 2024; Shen et al., 2025; Kim et al., 2025). For presentation authoring, a deck-wide author retains responsibility for all cross-page decisions, while per-page authors must coordinate choices that remain coupled (Figure 1). Sharing established facts and

plans therefore leaves a separate question: *which remaining cross-page adjustment needs should determine responsibility boundaries?*

PPTAgent adopts a two-stage, edit-based approach to presentation generation (Zheng et al., 2025). DeepPresenter separates research from deck-wide production through Researcher and Presenter roles, using a shared environment, rendered feedback, and trajectory training (Zheng et al., 2026). Within production, we study which pages should share an Owner. Our organizing principle distinguishes established information that can be shared from choices that still require joint adjustment. Pages can reuse a chart independently when its geometry is fixed, while successive reveals may require joint authorship when that geometry must be adjusted across the sequence. Shared style or evidence alone therefore does not justify grouping.

MURALPRESENTER uses shared deck state to connect local production with continuing responsibility. Owners retrieve relevant facts, plans, design rules, and assets without inheriting the full upstream interaction history. Single Owners handle independent pages, while Group Owners handle page sets with recorded joint-adjustment goals. They author, inspect, and revise their assigned units, with responsibility preserved when execution context must be reconstructed. The Orchestrator reviews the deck, coordinates changes spanning units, and routes revisions to the assigned Owners. We support this workflow by training 9B and 27B models on 9,000 quality-filtered execution segments.

Task-specific rubrics assess the quality of delivered slides (Chen et al., 2026), while long-horizon agent evaluations also examine intermediate artifacts using expert criteria (Chandwani and Gupta, 2026). We introduce **THREADEVAL**, a fine-grained, evidence-grounded protocol for examining requirement preservation during authoring and fulfillment in the final deck. The protocol links page- and object-level judgments to task-specific criteria and supporting evidence. Process assesses intermediate execution and handoffs separately from Final, which assesses delivered knowledge and presentation quality. The Knowledge checklist covers explicit content obligations and task-authorized depth, application, and synthesis.

In summary, our contributions are threefold:

- We propose **MURALPRESENTER**, a skill-guided multi-agent framework that combines persistent deck state with Single or Group ownership according to remaining cross-slide adjustment needs, maintaining responsibility through authoring, inspection, and revision.
- We introduce **THREADEVAL**, a fine-grained, evidence-grounded evaluation protocol that connects task-specific requirements with page- and object-level evidence, assessing intermediate execution separately from final presentation quality.
- We train 9B and 27B variants on 9,000 quality-filtered execution segments using selective action supervision and examine responsibility and supervision choices. Matched comparisons show that local ownership improves final quality with greater recorded workload, while Global Selective records fewer input tokens and agent turns on selected completed runs.

2 MURALPRESENTER

In this section, we present MURALPRESENTER, a multi-agent framework for long-horizon presentation authoring. We describe its shared-state orchestration and the assignment of Single or Group Owners based on whether pages require joint adjustment (Figure 2).

2.1 Task Definition

Given a user request q and optional source materials $\mathcal{M}$, the task is to produce an ordered, editable HTML deck $D = (s_1, \ldots, s_N)$, with page renders and per-page speaker notes. Authoring requires coordinating claim selection and ordering, supporting visuals, and page layouts for the intended audience. Narrative guides evidence placement, while layout constrains how it can be explained. These decisions must preserve factual conditions, uncertainty, and cross-page relations; subsequent revisions incorporate new user requirements into the existing deck.

2.2 Multi-Agent Collaboration

The Orchestrator coordinates Research, Image, and Slide subagents through a shared workspace.

Orchestrator. Maintains the deck plan and shared decisions, delegates tasks as their required inputs become available, and coordinates cross-unit revisions.

Research & Image. Research gathers and synthesizes evidence for delegated questions, linking claims to source locations and preserving conditions and uncertainty. Image prepares visual assets

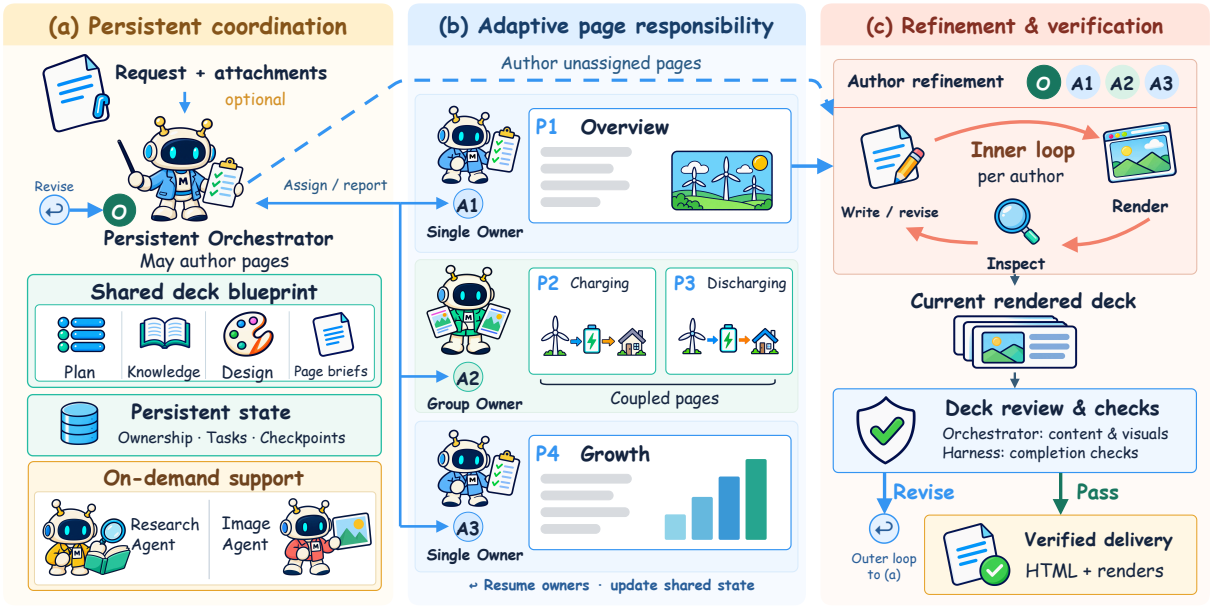


Figure 2: **Long-horizon authoring in MURALPRESENTER.** Shared state carries established decisions into local production; Owners refine assigned pages and the Orchestrator verifies the full deck. Content revisions resume existing Owners. "Regroup" applies only to structural revisions; shared-style changes use shared transactions.

and hands off their paths, intended uses, and limitations.

Slide. Turns page briefs and shared inputs into HTML pages and speaker notes, then inspects current renders and revises defects. Slide agents serve as Single or Group Owners, as defined below.

A *Skill* provides workflow and role instructions; the *Harness* enforces write scopes and schedules non-conflicting tasks in parallel. Appendix D provides role-instruction excerpts.

Production alternates local refinement with deck-level review. Owners inspect current page renders, and Group Owners also inspect relations within their units. The Orchestrator reviews contact sheets and targeted pages across units, then routes revisions to the responsible Owners. Acceptance requires current inspection and resolution of registered blocking issues; visual and semantic judgments remain fallible (Huang et al., 2024; Chen et al., 2024a).

2.3 Unit-Adaptive Responsibility Assignment

MURALPRESENTER introduces parallel slide production with unit-based ownership. Multiple Slide subagents concurrently author disjoint single pages or coupled page groups as *Single Owners* or *Group Owners*, retaining responsibility through inspection and revision. This complements functional role decomposition in prior presentation agents (Zheng et al., 2026; Liang et al., 2025) with an explicit page-to-Owner assignment. The pages form a partition $\mathcal{P} = \{U_1, \ldots, U_m\}$, with one Owner $\rho(U_j) = o_j$ per unit. Owners read relevant shared state $\mathcal{S}$: constraints, adopted facts and uncertainty, deck/page plans, design rules, assets, and inspection evidence.

For each ready page brief, the Orchestrator asks whether it can be completed independently *after shared evidence, design rules, assets, and coordinates are fixed*. If so, it assigns a Single Owner; otherwise, it forms a Group with an explicit joint-adjustment goal. Shared topic, style, or adjacency alone does not justify grouping. For example, successive chart reveals require a Group when their axes, geometry, or encodings still need joint adjustment; a fixed base chart can be reused by independent Single Owners.

Assignments follow an incremental, Skill-guided heuristic rather than an optimization over group size. Page scopes must be disjoint and registered before execution. Ordinary edits return to the assigned Owners; the Orchestrator coordinates new cross-unit coupling and shared-style changes. Changing page membership requires an explicit structural revision of plans, page relations, and Owner records.

Responsibility persists when local context must be recovered. An Owner may `Resume` compatible history, `Rebuild` missing image context, or start `Cold` from current artifacts without implicitly changing responsibility. Persistent responsibility therefore does not require an intact transcript. Ap-

Dimension	Category	Count	Share (%)
Language	English	407	27.32
	Chinese	1,083	72.68
Query type	Domain	792	53.15
	Simple Persona	299	20.07
	Elite Persona	199	13.36
	Attachment	200	13.42
Target length	Short ($\leq$ 10)	81	5.44
	Medium (11–20)	344	23.09
	Long ($>$ 20)	989	66.38
	Free	76	5.10
Total		**1,490**	**100.00**

Table 1: Composition of 1,490 candidate queries before filtering. Length is in pages; Free denotes no fixed count.

pendix E.2 details write scopes, continuation, and inspection receipts.

3 Trajectory Training

Our training pipeline comprises three stages: constructing presentation requests, synthesizing authoring trajectories, and filtering execution segments. Filtering assesses source-deck quality separately from local action eligibility.

3.1 Query Construction

GPT-5.5 constructs requests from domain scenarios, PersonaHub profiles (Ge et al., 2024) paired with O*NET tasks, and page-indexed PDFs, with content, visual, and cross-page requirements. Table 1 summarizes 1,490 candidate queries: 748 native requests and 742 archived domain requests.

3.2 Trajectory Synthesis

Teacher agents execute the workflow in Section 2, retaining tool interactions, renders, revisions, and images. Segments follow local agent/context boundaries: history remains input; newly generated assistant actions form targets. Teacher mappings appear in Appendix B.2.

3.3 Trajectory Filtering

Deterministic checks verify role/tool flow, artifacts, constraints, and HTML–render correspondence. Gemini 3.5 Flash assesses task fulfillment, factual consistency, and visual quality. Hard failures and unresolved high-severity findings block deck admission. A text-only DeepSeek-v4-pro judge separately annotates action eligibility; eligible actions alone do not establish source-deck quality.

From 748 native source records, we retain **9,000 execution segments from 686 sources**, excluding

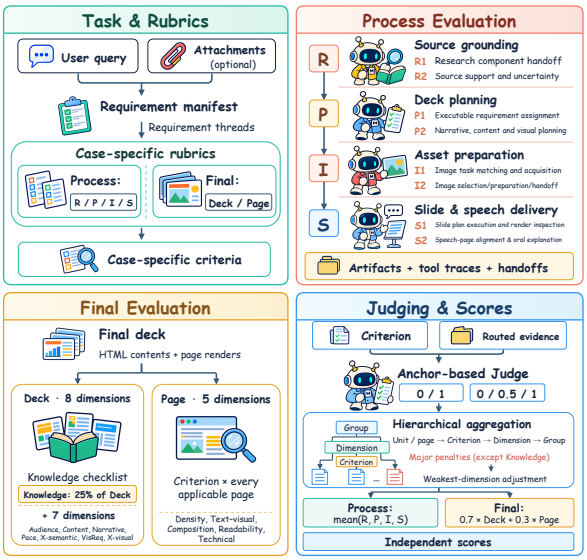


Figure 3: **Overview of THREADEVAL.** Task-specific criteria evaluate evidence-grounded process handoffs across R/P/I/S and final Deck/Page quality. Criterion–unit judgments use predefined anchors.

historical imports and artifact-first recovery. Appendix B reports role counts and admission details.

Training objective. Global Selective supervises current targets while masking historical assistant loss. We update the language backbone with frozen vision components and retain epochs two and four. Policy comparisons and mask details appear in Section 5.3 and Appendix B.5.1.

4 THREADEVAL

To assess requirement fulfillment during authoring and final presentation quality, we introduce THREADEVAL, which reports separate **Process** and **Final** scores. Requirement threads link upstream support, intermediate handoffs, and final realization (Figure 3).

4.1 Tasks and Rubrics

Our 35 Chinese/English tasks span six domains and request 20–40 pages using open research, supplied attachments, or both. Instructions specify audiences, time frames, source-use rules, and delivery constraints. Task-specific Process and Final rubrics declare requirements, applicable units, permitted evidence, and scoring anchors.

Deck's **Knowledge checklist** distinguishes *explicit content requirements* from *query-authorized knowledge*: depth, application, and synthesis warranted by the request (Sharma et al., 2026). Reference answers, accepted equivalents, and uncertainty boundaries are fixed before scoring. Optional exposition creates no additional obligation.

Construction, statistics, and a task–rubric example appear in Appendices C.2, C.3.1, and C.3.2.

4.2 Evaluation Protocol

Process evaluates source grounding and research-to-plan handoff (**R**), executable requirement assignment and planning (**P**), asset acquisition and handoff (**I**), and plan execution, render inspection, and slide–speech alignment (**S**) (Chandwani and Gupta, 2026). Judgments use intermediate artifacts, action records, and relevant delivered pages and notes.

Final assesses textual content from extracted final HTML and visual quality from page renders. **Deck** covers Knowledge, content selection, narrative organization and pacing, audience fit, cross-page semantic and visual consistency, and specified visual requirements. **Page** covers information hierarchy, text–visual correspondence, composition, readability, and technical integrity.

Final quality cannot establish an earlier handoff or inspection (Chen et al., 2026; Yang et al., 2026). Systems without compatible process-evidence adapters report Final and evidence coverage rather than a Process ranking (Appendix C.5).

4.3 Scoring and Aggregation

Criterion–unit judgments use declared $0/1$ or $0/0.5/1$ anchors; deterministic checks run in code. Archived Process/Final evaluations primarily use Gemini 3.5 Flash. Strict Final-only comparisons use DeepSeek V4.1 Flash for textual criteria and Gemini 3.5 Flash for visual criteria (Appendix C.1).

Applicable unit scores are averaged within criteria, then within dimensions. Capped deductions penalize predeclared major failures, except in Knowledge. A weakest-dimension adjustment applies once per group:

$$G_{\text{eff}} = G_{\text{base}}[1 - 0.2(1 - G_{\min})]. \qquad (1)$$

Here, G_{base} is the applicable weighted mean and $G_{\min}$ the minimum penalty-adjusted dimension score; group scores lie in $[0, 1]$. Knowledge receives 25% of Deck weight. Final combines adjusted Deck/Page at 70:30; Process averages applicable R/P/I/S groups. Their equal mean is the auxiliary Overall. Required omissions receive zero; evaluator failures block a valid aggregate. Tasks receive equal weight. Complete weights and deduction rules appear in Appendix C.6; aggregation sensitivity is examined in Section 6.

5 Experiments

We evaluate presentation quality, execution, and editing, then compare responsibility and supervision strategies. Section 6 analyzes their quality–workload trade-offs.

5.1 Setup

Implementation details. We fine-tune Qwen-3.5-9B/Qwen-3.8-27B on 9,000 segments from 686 native sources (Section 3), using MS-SWIFT Megatron/BF16 with frozen vision encoders and mergers. Training uses a 131,072-token limit, batch size 64, four epochs, peak learning rates $5 \times 10^{-6}/3 \times 10^{-6}$, and 8/32 GPUs. Main Global Selective checkpoints are 9B ep4/128 and 27B ep2/64, distinct from the ep4 supervision study (Appendix B).

Models and baselines. Untuned 9B/27B controls and seven THREADEVAL reference backbones use our runtime; external systems retain published protocols. Evaluations cover 238 PresentBench, 189 SlidesGen-Bench, 35 THREADEVAL, and 294 five-turn DECKBench tasks (Jang et al., 2026). Reproduction materials appear in Appendix E.4. The pending Codex + Vanilla PresentBench comparison requires aligned inputs, backbone, tools, budgets, and selection rules.

Comparison scope. Ten PresentBench and ten THREADEVAL tasks informed checkpoint selection and remain in reported means; semantic training–evaluation disjointness is unverified. PresentBench's 72.6 includes score-conditioned selection, preventing matched-budget SFT interpretation. The THREADEVAL archive mixes formal/diagnostic results, unlike strict Final-only studies. Section 4 defines metrics; Appendices C.1 and A.7.1 specify inclusion.

5.2 Main Results

Presentation quality. On SlidesGen-Bench, MURALPRESENTER-27B leads listed Content (94.92%) and Engagement (8.42); 9B's 90.21% Content exceeds all listed public systems (Table 3). Content gains over untuned backbones are 1.59/15.50 percentage points for 27B/9B.

Aesthetics gains are 3.92/0.54 for 9B/27B, but neither leads total Aesthetics and 27B Harmony declines. PresentBench 9B gains 6.21 Overall points; selected 27B's 72.6 leads listed Overall (Table 2), but score-conditioned selection prevents fixed-budget training interpretation.

Execution and delivery. Qwen-3.8-Max leads

Model	**Overall** ↑	Fund. ↑	Vis.&Lay. ↑	Compl. ↑	Corr. ↑	Fidelity ↑
Reported results						
hwc-mmi-aippt	<u>70.8</u>	90.3	61.1	**79.9**	**72.2**	50.3
NotebookLM	62.5	81.0	62.8	67.8	56.0	45.1
Manus 1.6	57.8	80.1	53.7	63.6	46.2	45.4
Tiangong	54.7	77.7	47.2	<u>68.8</u>	45.7	34.3
Zhipu	53.6	73.3	40.6	63.0	47.1	44.1
PPTAgent v2	50.2	79.8	44.4	60.2	37.9	28.8
Gamma	49.2	66.9	22.6	54.3	47.7	54.1
Doubao	48.0	71.8	40.7	58.2	34.7	34.8
Qwen	35.9	53.1	21.9	29.7	29.9	44.6
MURALPRESENTER runtime						
Qwen-3.5-9B	41.1	67.4	22.6	59.1	36.8	19.7
Qwen-3.8-27B	69.4	<u>94.4</u>	<u>70.9</u>	63.8	62.1	<u>55.8</u>
MuralPresenter-9B	47.3	74.4	43.1	50.3	40.4	28.5
MuralPresenter-27B	**72.6**	**94.6**	**73.5**	67.5	<u>66.6</u>	**60.8**

Table 2: PresentBench: 238 tasks per runtime row. Fund./Vis.&Lay. are material-independent; other dimensions are material-dependent. Bold/underline mark the best/second-best listed values throughout the main tables, without harmonizing source protocols. Settings: Appendix C.1.

System	**Content** (%) ↑	Aesthetics ↑				
		Total	Usability	Engagement	Harmony	Rhythm
Reported results						
Zhipu	88.29	22.06	4.87	7.52	-1.60	11.27
Kimi-Banana	87.09	<u>26.58</u>	**5.72**	6.41	-0.55	**15.01**
Skywork-Banana	83.83	**27.28**	<u>5.62</u>	<u>8.30</u>	<u>-0.47</u>	<u>13.84</u>
Kimi-Smart	82.07	18.30	4.13	7.99	-1.88	8.06
Quark	81.40	16.86	5.03	7.41	-1.91	6.33
Skywork	81.29	20.69	4.83	7.60	-1.18	9.44
Kimi-Standard	79.92	19.25	4.61	6.28	-1.75	10.12
NotebookLM	74.21	22.82	4.13	7.32	**-0.35**	11.72
Gamma	70.32	17.09	5.31	6.31	-1.51	6.99
MURALPRESENTER runtime						
Qwen-3.5-9B	74.71	14.46	4.22	8.25	-2.47	4.45
Qwen-3.8-27B	<u>93.33</u>	18.58	4.64	8.07	-1.35	7.22
MuralPresenter-9B	90.21	18.38	4.78	8.24	-2.14	7.51
MuralPresenter-27B	**94.92**	19.12	4.78	**8.42**	-1.45	7.37

Table 3: SlidesGen-Bench QuizBank accuracy and computational aesthetics. Ours: macro-means over 189 tasks; public protocols follow Yang et al. (2026). Total sums unrounded components. Details: Appendix C.9.2.

archived THREADEVAL Final (83.35; Table 4). Selected 9B gains 2.78 Final/9.13 Process points over its untuned backbone, despite a 3.38 Page decline. Selected 27B gains 2.05 Process but loses 0.26 Final. Section 6 examines their subdimension profiles; Appendix A.7.1 documents archive inclusion.

Multi-turn editing. MURALPRESENTER-27B completes five-turn edits on all 294 DECKBench tasks (Table 5). Its positive relative similarity gains remain below the official GPT-5-mini Medium pipeline with the same simulator and granularity. The matched trained–untuned intervals include zero for both metrics. Each turn imports preceding artifacts into a fresh workspace, testing artifact continuity rather than persistent Owner history (Appendix A.6).

Model	**Final** ↑	**Process** ↑	Overall† ↑
Claude Opus 4.7	77.86	79.31	<u>78.58</u>
Claude Sonnet 4.6	75.52	78.76	77.14
GPT-5.6 Sol	<u>80.12</u>	74.99	77.55
GPT-5.6 Terra	75.84	68.68	72.26
Gemini 3.5 Flash	75.14	58.98	67.06
Kimi K2.6	68.49	67.88	68.19
Qwen-3.8-Max	**83.35**	**80.32**	**81.83**
Qwen-3.5-9B	56.25	45.88	51.07
Qwen-3.8-27B	76.71	78.07	77.39
MuralPresenter-9B	59.03	55.01	57.02
MuralPresenter-27B	76.45	<u>80.12</u>	78.29

Table 4: THREADEVAL descriptive archive (35 tasks). †Overall averages Final and Process. Breakdowns and coverage: Appendices A.7 and A.7.1.

Editor	Simulator	ΔDTW ↑	ΔTransSim ↑
Reported results			
GPT-5-mini	GPT-5.1	**16.82**	**21.45**
DeepSeek-V3.1	Kimi K2	2.49	6.15
MURALPRESENTER runtime			
Qwen-3.8-27B	GPT-5.1	3.95	<u>20.68</u>
MuralPresenter-27B	GPT-5.1	<u>4.51</u>	18.34

Table 5: DECKBench editing (Medium; five turns): mean relative similarity changes, reported in units of 10^{-3}. Official pipelines follow Jang et al. (2026); Qwen-3.8-27B and MURALPRESENTER share our runtime. Full granularity results and paired intervals: Appendix A.6.

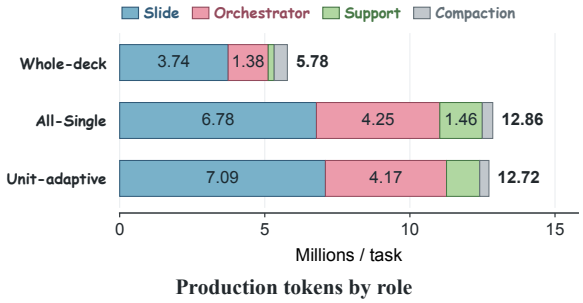


Figure 4: Mean production tokens by role (27B ep2/64; 35 tasks): recorded input plus output, excluding shared preparation. Support combines Research and Image; Compaction denotes context summarization. Totals are lower bounds.

5.3 Ablation Studies

Responsibility granularity. On 35 matched tasks, 27B Global Selective ep2/64 uses one deck-wide Owner (Whole-deck), one per page (All-Single), or Single/Group Owners (Unit-adaptive). Preparation, coordination, support roles, tools, and verification remain fixed. Whole-deck is an internal control rather than DeepPresenter. Table 6 details quality; Figure 4 breaks down production tokens by role, excluding shared preparation (Appendix A.2).

All-Single/Unit-adaptive improve Final over Whole-deck by +2.40 [0.45, 4.29]/+2.49 [0.83, 4.14] (paired 95% CIs), with greater tokens and turns. Adaptive versus All-Single is +0.09 [−1.35, 1.60], with uncertain workload reductions. Local responsibility improves quality; an additional grouping advantage remains unresolved.

Supervision. Full supervises historical/current actions; Global selects current targets across roles; Role-wise selects only for Slide. Trained arms share segments, ordering, packing, and 27B ep4/128 settings. All four arms pass strict Final-only scoring on 35 common tasks with identical references. Historical search, image backends, and quotas differ, preventing a pure training-effect interpretation (Appendix A.3).

Global has the highest Final (83.86; Table 7): paired differences are +4.02 [2.21, 6.03] versus Qwen-3.8-27B, +2.20 [−0.22, 4.57] versus Full, and +3.29 [0.92, 5.73] versus Role-wise. The Full contrast remains uncertain; intervals quantify task sampling rather than historical generation confounding.

Global records 15.50M input tokens versus Full's 25.01M (−38.0%; Figure 5(b)) and 455 versus 652 turns (−30.2%). Output changes by an uncertain +4.5%. Global exceeds Qwen-3.8-27B's workload: 15.87M/455 versus 11.16M/342 (tokens/turns). Completed-chain counts include compaction/continuations, excluding campaign cost and elapsed time.

6 Analysis

6.1 Responsibility and Coordination

Adaptive groups 139/961 pages across 16 tasks, with 17/2/16 wins/ties/losses against All-Single (Appendix A.2). Groups are not independently annotated high-coupling cases. Preparation may resolve joint choices; context, parallelism, and coordination vary together, preventing isolation of grouping.

6.2 Selective Supervision and Execution

Full/Global/Role-wise supervise 36.73M/14.87M/19.13M token positions per corpus pass (Table 23). Full supervises shared Global target calls 2.02 times versus once in Global. Selection changes repetition and action coverage; per-token normalization changes role label mass. Reduced historical-action weighting is a plausible explanation, not an isolated causal finding.

Slide calls fall from 346 to 247 per task; median input falls only from 26.9K to 25.2K (Table 13). Input savings accompany fewer calls rather than proportionally smaller contexts; call counts cannot identify redundant reading, planning, or inspection.

6.3 Requirement Fulfillment and Delivery

Archived 27B handoff/acquisition scores rise from 78.27/81.67 to 87.94/88.26, while S1 realization/inspection and content coverage fall from 89.26/96.44 to 76.03/88.96 (Tables 19–20). Upstream gains coexist with weaker delivery; diagnosing lost requirements needs aligned evidence.

Configuration	Deck-level quality					Page-level quality		Summary	
	Knowl.	Content	Narrative	Semantic	Visual	Layout	Readab.	Final	Δ
Whole-deck	50.96	89.38	**89.81**	**89.38**	**94.17**	99.47	97.97	77.65	–
All-Single	**54.96**	92.69	87.57	**89.38**	86.52	99.91	98.54	80.05	+2.40
Unit-adaptive	53.76	**93.74**	**89.81**	89.19	87.23	**99.94**	**99.80**	**80.14**	**+2.49**

Table 6: Responsibility-policy quality comparison (27B ep2/64; 35 matched tasks). Task means on a 0–100 scale. Knowl.: Knowledge; Readab.: font/color readability; Semantic/Visual: cross-slide consistency. Final uses the full official aggregation; Δ is its change relative to Whole-deck before rounding.

Configuration	Final ↑	Δ	Deck ↑	Page ↑
Qwen-3.8-27B	79.83	–	73.01	95.77
Full	81.66	+1.82	74.92	97.38
Global Selective	**83.86**	**+4.02**	**77.92**	**97.71**
Role-wise Selective	80.57	+0.73	75.29	92.88

Table 7: Four-arm 27B Final-only comparison (35 tasks; SFT: ep4/128). Δ denotes unrounded Final differences from Qwen-3.8-27B. Comparison scope: Appendix A.3.

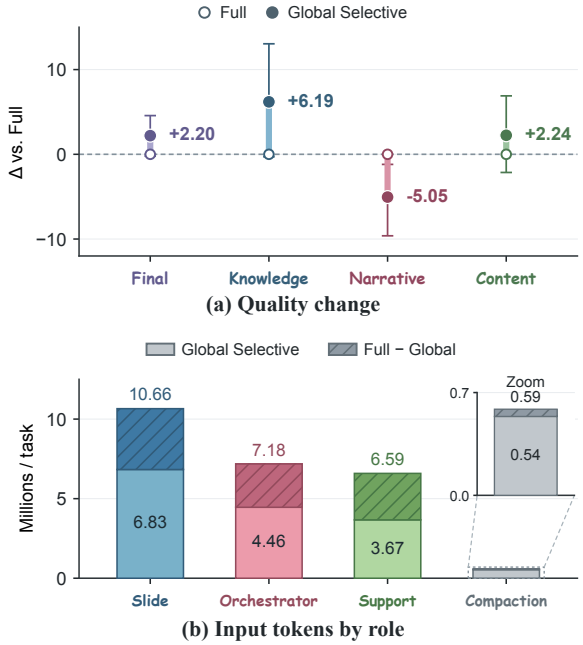


Figure 5: Global Selective versus Full (27B ep4/128; 35 tasks). (a) Global-minus-Full quality differences with paired 95% bootstrap CIs; dimensions are exploratory. (b) Input tokens on selected completed chains: Global bases plus Full's excess (hatched). Support combines Research and Image.

On 27 tasks containing both Knowledge types, Global improves explicit-content satisfaction by +9.59 [3.19, 16.68], versus uncertain query-authorized gains of +2.03 [−6.74, 12.28] (Appendix A.5). These diagnostics do not test whether effects differ or replace criterion-weighted Knowledge. Aggregate Knowledge gains (+6.19) remain uncertain, while Narrative falls by 5.05 (Figure 5).

Aggregation sensitivity. Global > Full > Role-wise persists across four aggregation rules (gaps 1.11–2.20) and fixed length strata (Tables 15 and 16). Adjustments change magnitude, not ordering; robustness establishes neither length-dependent benefit nor human–Judge agreement.

7 Related Work

Presentation agents. AutoPresent, PPTAgent, and PreGenie support programmatic authoring and rendered-artifact revision (Ge et al., 2025; Zheng et al., 2025; Xu et al., 2025). SlideBot centralizes coordination; SlideGen and DeepSlides separate production stages (Xie et al., 2026; Liang et al., 2025; Cui et al., 2026). DeepPresenter already combines Researcher–Presenter collaboration, a shared environment, render-grounded reflection, and verified trajectory training (Zheng et al., 2026). Our distinction is production responsibility after information sharing: when pages can be independent, when joint adjustment remains, and how the resulting scopes continue into revision. MemSlides complements this question with stateful edit-scope preservation (Jin et al., 2026).

Execution and evaluation. External state, context folding, and agent topologies manage growing histories and collaboration (Liu et al., 2024; Sun et al., 2025; Zhu et al., 2026; Shen et al., 2025); trajectory supervision supports tool use and repair (Zeng et al., 2024; Chen et al., 2024b). PresentBench and SlidesGen-Bench assess delivered presentations, and editing benchmarks assess requested changes (Chen et al., 2026; Yang et al., 2026; Jang et al., 2026). Following evidence-based process evaluation (Chandwani and Gupta, 2026), THREADEVAL distinguishes observable handoff quality from final realization. It does not equate an unavailable log with an action never occurring.

8 Conclusion

MURALPRESENTER defines production responsibility after fixed shared decisions and separates handoff evidence from delivery. Matched local

responsibility improves Final over Whole-deck at greater workload, without established adaptive gains over All-Single. Descriptively, Global exceeds Vanilla in Final; its Full contrast remains uncertain. Lower cumulative input and turns than Full accompany fewer calls, while workload exceeds Vanilla. These trade-offs motivate controlled tests of grouping and context continuity.

Limitations

The responsibility comparison fixes prepared inputs and tests production granularity; it does not isolate grouping rationale or context continuity. Supervision results describe historically generated artifacts with differing tool conditions, not a causal training effect. PresentBench's selected 27B score includes score-conditioned regeneration. Checkpoint-screening tasks remain in reported means, selection membership is not fully reconstructed, and semantic training–evaluation overlap is unverified. Workload covers selected completed chains rather than full experiment cost or elapsed time.

THREADEVAL contains 35 tasks and measures observable evidence. Missing logs do not prove missing work, and partial candidate-page retrieval can omit relevant evidence. No independent human–Judge calibration or human routing audit has been completed. Near-ceiling Page scores limit discrimination; the historical archive includes diagnostic evaluations and differs from strict scoring. Aggregation and length checks assess stability of the trained-arm ordering, not measurement validity or Process-ranking robustness.

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A Additional Experiments and Analyses

A.1 Experimental Configurations and Comparison Scope

Reference systems. THREADEVAL compares seven general-purpose reference backbones, both untuned backbones, and the two MURALPRESENTER models under the authoring runtime. PresentBench and SlidesGen-Bench comparisons use the untuned and trained 9B and 27B systems. Public scores retain their source protocols in separate *reported* blocks and are not pooled with our runs. Existing outputs are reused only when cases and run configurations match; incomplete results remain pending. Process comparisons require compatible evidence adapters held fixed across systems (Appendix C.5).

A.2 Responsibility Granularity

Responsibility controls. The comparison fixes prepared research, page plans, design rules, available assets, page counts, and verified input hashes for the same 35 tasks, using 27B Global Selective ep2/64 under fixed authoring instructions and runtime code. Whole-deck assigns one persistent Slide Owner to all pages, while retaining the Orchestrator and support roles; it is neither a single-agent replacement of the system nor an implementation of DeepPresenter. It controls ownership granularity within MURALPRESENTER, while DeepPresenter is a separate authoring system. All-Single enforces one Slide Owner per page and still permits shared coordination. Unit-adaptive permits Single and Group Owners according to unresolved joint-production needs. Revisions return to the existing Owner scopes. The comparison varies context size, parallelism, and coordination together and does not isolate the grouping criterion.

All arms deliver the prescribed 961 pages across 35 tasks, and all 105 outputs pass strict Final-only scoring and native-artifact validation. The same frozen case rubrics and Judge-only references are used across arms, with DeepSeek V4.1 Flash for textual criteria and Gemini 3.5 Flash for visual criteria. Reference omissions are repaired consistently across arms rather than treated as generation failures. The first strictly valid score is retained; invalid judgments do not enter the aggregate. Text judging uses the declared same-version model through TokenHub and the official DeepSeek route, with routing recorded; independently identical provider weights are not attested.

Assignment	Final	Tokens (M)	Turns
Whole-deck	77.65	**5.78**	**114**
All-Single	<u>80.05</u>	12.86	388
Unit-adaptive	**80.14**	<u>12.72</u>	<u>371</u>

Table 8: Responsibility on 35 matched tasks (27B ep2/64). Final: strict quality; tokens/turns: recorded production workload, excluding preparation. Tokens are lower bounds. Controls and paired intervals: Appendix A.2.

Process scores are not endpoints.

Workload excludes inherited preparation attempt IDs and deduplicates production attempts. Tokens include recorded input plus output for all roles, compaction, and continuations; cached input is not counted twice. Turns count completed author-model responses, with compaction calls accounted for separately. All successful responses have reported usage. Failed provider attempts without usage number 0, 35, and 19 for Whole-deck, All-Single, and Unit-adaptive, respectively, so recorded token totals are lower bounds rather than exact billing. The comparison conditions on the same prepared inputs and evaluates completed production chains rather than end-to-end reliability. Recorded workload does not measure total experiment cost or wall-clock speedup.

Unit-adaptive uses a Group Owner on 16/35 tasks, jointly owning 139/961 pages. Mean Slide Owner counts are 1.00 (Whole-deck), 27.46 (All-Single), and 24.60 (Unit-adaptive); realized partitions are diagnostics, not outcome-based labels of which tasks require grouping. Adaptive wins/ties/loses on Final in 17/2/16 tasks against All-Single and 24/0/11 against Whole-deck. No pre-annotated dependency-stratified effect or matched alternative grouping is estimated. Ten tasks informed checkpoint selection elsewhere, so this 35-task policy comparison is not a wholly held-out estimate. A dependency-cut or joint-requirement satisfaction score requires additional annotations and is not a current THREADEVAL endpoint.

A.3 Supervision Strategies

Supervision controls. Full, Global Selective, and Role-wise Selective use the same 9,000 records, ordering, packing, and 27B ep4/128 training settings. Generation records are selected by completion rather than quality score. All 105 outputs cover the same 35 tasks, runtime release, Skill, and medium reasoning setting; native HTML/render hashes and selected continuation traces are verified.

Paired contrast	Δ Final	Δ tokens (M)	Δ turns
Adaptive − All-Single	+0.09 [-1.35, 1.60]	-0.13 [-2.57, 1.81]	-16.83 [-63.63, 19.23]
Adaptive − Whole-deck	+2.49 [0.83, 4.14]	+6.95 [3.48, 12.56]	+256.43 [180.17, 384.72]
All-Single − Whole-deck	+2.40 [0.45, 4.29]	+7.08 [2.95, 14.30]	+273.26 [181.09, 436.17]

Table 9: Responsibility-policy differences on 35 matched tasks; brackets contain paired 95% percentile bootstrap intervals (10,000 task-level resamples, seed 20261003). Resampling preserves each three-arm task triple. These are unadjusted descriptive intervals from one completed production chain per arm and task, not independent page-level replicates. Tokens are recorded usage; provider calls missing usage are excluded rather than assigned zero.

Text criteria use DeepSeek V4.1 Flash and visual criteria Gemini 3.5 Flash, with fixed Final-only rubrics and aggregation. These are distinct from the main checkpoints (Appendix B.6); recorded workload does not measure elapsed time.

Qwen-3.8-27B baseline and comparison scope. Qwen-3.8-27B adds 35 strictly eligible Final-only results on the same requests and attachments, authoring instructions, runtime code, and evaluator code. All four arms now share per-task reference hashes and post-hoc native-artifact admission checks; Materials v014 alignment preserves every reported aggregate. Historical generation remains unmatched. Selected traces record unconfigured search in 10 Full, 7 Global, and 11 Role-wise outputs, and image requests sent to incompatible text endpoints in 9, 6, and 9 outputs, respectively. Qwen-3.8-27B records image quota failures in three outputs and uses a different image backend for 29 tasks. These observed tool conditions, together with unreconstructed historical budgets, prevent a pure training-effect interpretation; later scoring alignment does not remove them. Global minus Qwen-3.8-27B is $+4.02$ Final points (95% CI $[2.21, 6.03]$). Qwen-3.8-27B's completed chains average 11.16M tokens and 342 turns, versus Global's 15.87M and 455. No task is removed according to its score.

Aggregation and uncertainty. Tasks receive equal weight. The original trained-arm contrasts bootstrap 35 complete task triples 10,000 times (seed 20261002); comparisons with Qwen-3.8-27B use the same task-paired procedure. Percentile 95% intervals retain policy pairing and describe the observed completed runs. Figure 5(a) shows paired Global–Full quality differences with 95% intervals. Panel (b) decomposes mean recorded input tokens into Slide, Orchestrator, Support (Research plus Image), and Compaction. Its input ratio uses unrounded totals; completed-chain counts are not full-campaign costs. Role-level input accounting is reported in Table 12. Intervals condition on the selected runs, rather than training seeds or generation-run selection. The task set includes checkpoint-screening cases, and semantic training–evaluation disjointness is not established; these intervals do not quantify held-out generalization.

Scoring validity. All 140 four-arm scores pass strict completion, eligibility, model-routing, page-coverage, and official-aggregation checks; native HTML, page renders, and instructions match their pre-scoring hashes. The same post-hoc admission policy verifies canonical page pairing and file integrity for every artifact. Reference catalogs use official NIST Chemistry WebBook and RFC 9849 sources, with fixed criteria, anchors, and identical reference files across all four arms for each task.

Task-paired quality results. Table 10 gives all aggregate means and paired differences, including the completed descriptive Qwen-3.8-27B baseline. Intervals are not adjusted for multiple comparisons; near-ceiling Page scores limit discrimination. No prespecified non-inferiority margin was tested.

Dimension-level diagnostics. Table 11 reports all 13 official dimensions with 35/35 case coverage for each of the three policies. These unadjusted, post-hoc diagnostics describe where observed differences occur; they do not establish an overall improvement in every aspect or a multiplicity-controlled dimension effect.

A.4 Resource Usage and Context Analysis

Efficiency accounting. Resource statistics sum recorded input and output tokens across all roles, compactions, and continuations within completed execution chains. Cached input is a subset of input usage and is not added again; totals count token use across requests rather than unique context content. Agent turns count completed work rounds; compaction calls are separate. All 105 analyzed chains have complete traces, but these counts do not measure total experiment cost or elapsed time. Unavailable usage is not treated as zero.

Configuration / paired contrast	Final [95% CI]	Deck [95% CI]	Page [95% CI]
Qwen-3.8-27B	79.83 [78.01,81.69]	73.01 [70.45,75.60]	95.77 [93.90,97.48]
Full	81.66 [79.44,83.81]	74.92 [71.87,77.93]	97.38 [96.00,98.58]
Global Selective	**83.86** [81.72,85.93]	**77.92** [75.04,80.75]	**97.71** [96.54,98.74]
Role-wise Selective	80.57 [77.68,83.36]	75.29 [71.76,78.82]	92.88 [86.35,97.48]
Global − Full	+2.20 [−0.22,4.57]	+3.00 [−0.44,6.34]	+0.33 [−1.27,1.89]
Role-wise − Full	−1.09 [−3.77,1.55]	+0.38 [−2.94,3.66]	−4.49 [−11.07,0.25]
Global − Role-wise	+3.29 [0.92,5.73]	+2.63 [0.11,5.24]	+4.83 [0.003,11.49]
Full − Qwen-3.8-27B	+1.82 [−0.26,3.87]	+1.91 [−1.12,4.95]	+1.60 [−0.15,3.57]
Global − Qwen-3.8-27B	+4.02 [2.21,6.03]	+4.92 [2.33,7.81]	+1.94 [−0.01,3.93]
Role-wise − Qwen-3.8-27B	+0.73 [−1.95,3.40]	+2.29 [−0.82,5.52]	−2.89 [−9.71,2.14]

Table 10: Strict Final-only aggregate scores on 35 common tasks (trained arms: ep4/128). Contrasts are descriptive score-point differences; comparison scope is defined in Appendix A.3. Intervals resample tasks rather than criterion judgments.

Official quality dimension	**Full**	**Global**	**Role-wise**	$\Delta_{\text{Global}-\text{Full}}$ [95% CI]
Knowledge checklist	55.14	**61.33**	54.00	$+6.19\ [-0.29, 13.04]$
Narrative structure / transition	**98.10**	93.05	93.62	$-5.05\ [-9.62, -1.19]$
Content selection / emphasis / visual coverage	95.37	**97.61**	96.88	$+2.24\ [-2.14, 6.90]$
Cross-page semantic consistency	87.19	89.05	**94.76**	$+1.86\ [-7.05, 11.43]$
Cross-page visual consistency	86.10	87.55	**87.69**	$+1.45\ [-6.40, 9.38]$
Text–visual relationship	98.77	**99.44**	94.31	$+0.68\ [-0.20, 1.60]$
Technical integrity / asset quality	94.56	**94.99**	89.70	$+0.42\ [-3.55, 4.49]$
Audience communication	93.10	**93.45**	92.86	$+0.36\ [-3.69, 4.29]$
Pacing / page allocation / redundancy	98.81	98.57	**99.05**	$-0.24\ [-2.38, 2.14]$
Visual requirement compliance	91.01	**91.24**	86.33	$+0.22\ [-4.16, 4.45]$
Information hierarchy / density	**99.94**	99.84	94.86	$-0.10\ [-0.32, 0.07]$
Composition / alignment / balance	99.84	**99.91**	97.75	$+0.07\ [-0.13, 0.29]$
Typography / color / readability	**99.77**	99.75	95.66	$-0.02\ [-0.20, 0.14]$

Table 11: Complete dimension-level training-strategy analysis on the same 35 tasks. Values are official pre-group-cap dimension scores (0–100), averaged equally across tasks; they are not interchangeable with the post-cap/penalty Deck, Page, or Final aggregates. Global denotes Global Selective. Differences and intervals use paired task bootstraps; intervals are unadjusted for 13 comparisons. Rows are ordered by absolute observed Global–Full differences, and the first three are displayed in Figure 5(a). This post-hoc selection is exploratory.

Relative to Full, Global Selective has paired mean differences of -9.50M tokens (95% interval $[-14.51, -4.45]$M) and -197.2 turns ($[-296.9, -94.4]$). Role-wise Selective differences are $+0.28$M tokens ($[-4.59, 5.13]$M) and $+15.9$ turns ($[-85.6, 120.4]$). The latter intervals include zero and do not establish equivalence.

The input decomposition in Figure 5(b) uses the same selected traces. Full and Global Selective consume 25.012M/15.497M input tokens and 0.357M/0.373M output tokens per task, respectively. Their Global/Full ratios are 61.96% (95% paired interval $[48.25, 79.92]$) for input, 104.50% ($[87.33, 124.74]$) for output, and 69.77% ($[57.65, 84.52]$) for turns. The paired input difference is -9.515M ($[-14.491, -4.516]$M); the output difference is $+0.016$M ($[-0.046, 0.089]$M). Thus, the observed token reduction is concentrated in input usage; output usage does not show a clear change. The ratios describe resource accounting, not a causal decomposition or elapsed-time speedup.

Role-level input burden. The 105 selected continuation chains have complete provider traces. Table 12 attributes input use to each role and compaction, counting cached input once. Slide, Orchestrator, and Research account for about 92% of the mean Global–Full input reduction. Provider-call means also fall from 345.9 to 247.1 for Slide, 137.5 to 92.4 for Orchestrator, and 81.5 to 42.1 for Research. These are provider requests with recorded usage, distinct from deduplicated agent turns. Role attribution describes where recorded work occurs; it does not label particular reading, checking, or planning actions as redundant.

Per-call context burden. Table 13 distinguishes cumulative input from the amount of context supplied to a single author-model call. We re-read all selected-chain provider traces for the same 105 runs, deduplicate attempt identifiers, and verify their input totals against the frozen resource au-

Input tokens / task (M)	Full	Global	Role-wise
Slide	10.66	**6.83**	9.85
Orchestrator	7.18	**4.46**	7.48
Research	3.28	**1.11**	5.21
Image	3.31	2.55	**2.15**
Compaction	0.59	**0.54**	0.55
Total	25.01	**15.50**	25.25

Table 12: Mean provider-recorded input use on 35 matched tasks. Totals are calculated before rounding. Accounting covers completed chains rather than total experiment cost or elapsed time.

dit. The table pools successful main author-model calls; it excludes compaction and does not treat calls as independent task replicates. Reported input includes cached tokens once. For Slide, Full-to-Global median context falls only from 26.9K to 25.2K, while calls per task fall from 346 to 247. Research changes more on both measures: median context falls from 36.8K to 22.6K and calls per task from 81.5 to 42.1. Thus lower cumulative input does not imply a proportionate reduction in every decision's context. This supervision comparison does not isolate an ownership-policy effect.

A.4.1 Selected 9B Resource Accounting

Selected 9B accounting. Separately from the original archive, the 35 selected 9B runs record means of 20.592M input and 0.616M output tokens, 461.8 model calls, 465.1 role-reported turns, and 920.6 tool calls. Usage for payment-terminal and plant-based-meat tasks is incomplete; tool histories are incomplete for ten tasks. Recorded usage does not establish full-campaign cost. Role-reported turns may include restored context, and the 66.54-hour sum of run durations is not batch elapsed time. Retained scoring, inventory, and HTML histories contain 29,648, 283, and 1,003 request attempts. Their estimated total is $862.77 at archived 26 September rates without cache discounts, not an actual bill or a fresh-evaluation budget. Operational records are provided with the reproduction material.

A.5 Robustness and Requirement-level Diagnostics

Requirement-type diagnostics. We reuse the strict ep4 judgments on the 27 tasks whose frozen checklists contain both explicit content requirements and query-authorized knowledge. For each type, criterion scores are averaged within a task and then equally across these 27 tasks. Expanded entity criteria retain their within-criterion means. The eight other tasks have no explicit Knowledge criteria; they are excluded from this paired type diagnostic, not assigned zero. These satisfaction rates are separate from the official criterion-averaged Knowledge aggregate: entity scores are averaged within each criterion, and applicable criteria are weighted equally; topics are not separately balanced.

Global–Full differences are $+9.59$ points (95% CI $[3.19, 16.68]$) for explicit content and $+2.03$ ($[-6.74, 12.28]$) for query-authorized knowledge. The latter remains uncertain. Intervals use 10,000 paired task resamples (seed 20261004) and are exploratory and unadjusted; they do not establish broad knowledge improvement.

Aggregation sensitivity. We reuse all 105 strict ep4 Final-only results and their original criterion judgments. The four variants fix dimension weights (Knowledge 25% of Deck) and Final's 70:30 Deck/Page weights, varying only major deductions and the weakest-dimension adjustment. Pre-deduction dimension values are retained in each score record; the official variant is reconstructed to within 0.0001 score points on every run. No new Judge calls are made. Table 15 reports case-balanced means. All four variants preserve the aggregate order Global Selective, Full, Role-wise Selective; the gap is not invariant. These 35-task means condition on selected completed runs and do not validate the main archive's Process ranking. The requirement-type and length-stratified diagnostics reuse the completed judgments and keep task membership fixed across policies.

Length-stratified ordering. Task membership is fixed using Full's delivered page count: 25 tasks have 20–28 pages and ten have 30–40. Each task's three policies remain in the same stratum, even when their realized page counts differ. All four aggregation variants retain Global $>$ Full $>$ Role-wise in both strata; Table 16 shows official Final means. This exploratory partition checks sensitivity of the observed ordering. It neither identifies the effect of deck length nor establishes increasing benefit on longer tasks.

A.6 Multi-turn Editing

Multi-turn editing. DECKBench Task 2 uses a GPT-5.1 user simulator with the balanced (Medium-granularity) persona and five editing turns. Qwen-3.8-27B and MURALPRESENTER-27B use the same frozen Skill/Harness and complete generation and scoring for the same 294 task

Policy	Role	Calls	Median (K)	P95 (K)	Maximum (K)
Full	Slide	12,108	26.9	59.3	88.8
	Orchestrator	4,811	50.7	83.2	102.1
	Research	2,852	36.8	90.2	117.0
	Image	3,126	35.3	67.7	82.6
Global Selective	Slide	8,649	25.2	50.7	84.4
	Orchestrator	3,234	46.9	78.7	109.4
	Research	1,473	22.6	56.1	114.8
	Image	2,579	33.0	62.7	81.8
Role-wise Selective	Slide	11,727	26.5	55.0	88.1
	Orchestrator	5,081	49.5	82.4	126.2
	Research	4,371	39.7	87.1	133.6
	Image	2,250	32.2	58.1	76.8

Table 13: Provider-recorded input tokens per successful author-model call in the 35 matched tasks per policy. K denotes 1,000 tokens. Calls are pooled within each role; percentile and maximum values describe these selected chains rather than full-campaign workload.

Knowledge requirement	Full	Global	Role-wise
Explicit content	79.87	**89.46**	82.93
Query-authorized	55.25	**57.28**	53.64

Table 14: Requirement-type satisfaction (0–100) on 27 matched tasks containing both types. All policies use identical frozen criteria; these diagnostic means do not replace official Knowledge scoring.

Aggregation	Full	Global	Role-wise	$\Delta_{\mathrm{G-F}}$
Mean only	88.35	**89.46**	87.31	+1.11
Major only	87.95	**89.16**	86.63	+1.21
Weakest only	82.30	**84.39**	81.51	+2.09
Both (official)	81.66	**83.86**	80.57	+2.20

Table 15: Final aggregation sensitivity on identical judgments. Global/Role-wise denote their Selective policies; differences are descriptive mean differences.

identifiers, yielding 1,470 edits each. Each starts from the benchmark's supplied deck and local assets; search and image generation are disabled. Each turn imports the preceding HTML and assets into a fresh workspace, so this evaluates artifact continuity rather than persistent Owner history. The qualitative restyling case in Appendix A.8 remains separate.

Table 5 follows the released analysis script's baseline-relative aggregation:

$$\Delta s = \frac{1}{N}\sum_{i=1}^{N}\frac{s_{i,5}-s_{i,0}}{|s_{i,0}|}, \quad N = 294, \qquad (2)$$

where s is DTW similarity or transition similarity to the reference deck. We recompute these means from all per-task turn-0/turn-5 scores; no denominator is zero. DTW measures sequence alignment, while TransSim compares adjacent-slide semantic transitions. The evaluator returns TransSim $= 1$ for sequences with fewer than two text-bearing slides to avoid undefined transitions. These metrics measure reference similarity relative to each initial deck; they are not absolute final-quality scores or direct measures of instruction adherence.

Full pages	Tasks	Full	Global	Role-wise
20–28	25	82.26	**84.00**	81.76
30–40	10	80.13	**83.51**	77.60

Table 16: Length-stratified official Final means on fixed matched task sets. Global and Role-wise denote their Selective policies.

Cross-method comparison. The main table compares MURALPRESENTER with DECKBench's official editing pipeline, retaining each method's execution framework. The GPT-5-mini Medium configuration shares our GPT-5.1 simulator and instruction granularity; the DeepSeek-V3.1 configurations instead use Kimi K2. This is a descriptive system comparison: observed gaps reflect the editor model together with its runtime, rather than an isolated backbone effect. Simulator and granularity settings remain explicit in both tables.

For comparison with unnormalized reports, mean raw differences $s_5 - s_0$ are 0.00297/0.01274 for untuned 27B and 0.00339/0.01194 for MURALPRESENTER-27B (DTW/TransSim). These differ from the relative changes in the main table. The main comparison consistently uses relative changes, and neither form measures total execution cost.

Paired training comparison. On the same 294 tasks, the trained-minus-untuned difference in per-deck relative improvement is $+0.00056$ for DTW (95% CI $[-0.00052, 0.00158]$) and -0.00234 for TransSim ($[-0.00985, 0.00455]$). Percentile in-

Editor	Gran.	ΔDTW	ΔTransSim
Reported results: GPT-5.1 simulator			
GPT-5-mini	Low	0.01006	0.01594
GPT-5-mini	Medium	0.01682	0.02145
GPT-5-mini	High	0.02278	0.03230
Reported results: Kimi K2 simulator			
DeepSeek-V3.1	Low	0.01101	-0.00147
DeepSeek-V3.1	Medium	0.00249	0.00615
DeepSeek-V3.1	High	0.01549	0.04287
MURALPRESENTER runtime: GPT-5.1 simulator			
Qwen-3.8-27B	Medium	0.00395	0.02068
MURALPRESENTER-27B	Medium	0.00451	0.01834

Table 17: Full DECKBench granularity breakdown. Changes are relative to each initial deck after five edits. Published rows retain their source settings (Jang et al., 2026); our two Medium rows form a matched training comparison within the same runtime.

tervals use 10,000 paired task resamples (seed 20261005), keeping both policies together, and are unadjusted for the two comparisons. Both include zero, so this study establishes no reliable mean improvement or decline. It does not test an equivalence or non-inferiority margin.

A.7 Detailed THREADEVAL Results

A.7.1 Archive Inclusion and Coverage

Result inclusion in the reported archive. The original eight reference-model rows in Table 4 average all 35 selected tasks per model, including 166 evaluations classified as formal and 114 as diagnostic across the 280 records. Diagnostic flags concern cross-evaluation reuse, accepted inventory-review limits, or accepted scoring fallbacks; flags can overlap and do not themselves denote generation failure. Thus, these are descriptive archive means, not uniformly strict-validation estimates. The formal/diagnostic counts, respectively, are 20/15 for Opus, 18/17 for Sonnet, 13/22 for GPT Sol, 26/9 for GPT Terra, 25/10 for Gemini, 26/9 for Kimi, 17/18 for Qwen-Max, and 21/14 for untuned Qwen-27B. The archived flags are retained rather than treating every completed evaluation as formal; filtering each model to its own formal subset would compare different task sets.

Additional archive rows. The selected 27B ep2/64 and untuned 9B each cover all 35 tasks with matching per-task rubric versions and the legacy presenter, retaining 11/24 and 9/26 formal/diagnostic records, respectively. Selected 27B has no scoring-fallback items; untuned 9B retains 42 across six tasks. These means follow the same inclusion rule; they are not a causal SFT contrast across different backbones.

Selected 9B evaluation. The selected 9B evaluation comprises 35 completed ep4/128 runs, selected by generation completion rather than score, with 13 formal and 22 diagnostic results. Nine tasks retain 12 accepted scoring-fallback items; materials inverse design and QUIC also retain missing-reference flags for unlisted objects. The selected and untuned 9B QUIC rubrics use different revisions. Excluding QUIC, the 34-task Final means are 59.14/56.34 and Process means are 55.18/45.61, respectively, preserving the direction of the full-set differences. The archive remains a descriptive comparison rather than a causal SFT estimate.

A.7.2 Complete Scores and Dimension Profiles

Table 18 retains the full grouped profile; Tables 19 and 20 give the subdimensions corresponding to Table 4. Selected 9B scores 59.45/83.60/43.48/33.51 on R/P/I/S: planning scores exceed image and slide/speech execution scores.

A.8 Qualitative Paper-to-Presentation Case Study

Figure A1 follows one paper-to-presentation workflow using MURALPRESENTER-27B (Global Selective, epoch 2). A blackboard-style deck was generated from this paper with search and image generation, expanded from ten to twelve slides, and then edited into a light blue-and-white presentation. The same authoring runtime retained the assigned Owners across revisions. Explicit feedback guided factual alignment, visual examples, and layout, with render inspection after edits. The paired decks illustrate content development and coordinated restyling of an existing artifact; they are a qualitative showcase rather than an estimate of average quality or a controlled editing result.

B Training Data and Training Details

B.1 Query and Attachment Construction

Native task construction. GPT-5.5 generates requests from domain, persona/occupation, and attachment conditions. Simple/Elite Persona tasks draw from PersonaHub's general/expertise pools (Ge et al., 2024) with matched O*NET occupations and tasks[1]. Heuristic domain weights and explicit visual directions diversify the requests.

[1] O*NET Database, version 30.3. https://www.onetcenter.org/.

Model	Overall	Final			Process				
		Score	Deck	Page	Score	R	P	I	S
Claude Opus 4.7	78.58	77.86	77.45	78.81	79.31	**73.30**	**97.94**	69.08	76.91
Claude Sonnet 4.6	77.14	75.52	72.99	81.43	78.76	68.04	93.77	73.11	80.11
GPT-5.6 Sol	77.55	80.12	79.59	81.35	74.99	72.47	96.65	75.57	55.27
GPT-5.6 Terra	72.26	75.84	73.61	81.05	68.68	63.66	83.62	77.18	50.26
Gemini 3.5 Flash	67.06	75.14	72.28	81.81	58.98	52.15	81.26	49.49	53.03
Kimi K2.6	68.19	68.49	64.23	78.42	67.88	58.79	85.05	63.67	64.01
Qwen-3.8-Max	**81.83**	**83.35**	**80.35**	**90.35**	**80.32**	65.66	96.92	74.92	**83.77**
Qwen-3.5-9B	51.07	56.25	53.27	63.21	45.88	47.88	63.44	43.30	28.90
MURALPRESENTER-9B	57.02	59.03	58.69	59.83	55.01	59.45	83.60	43.48	33.51
Qwen-3.8-27B	77.39	76.71	72.38	86.81	78.07	65.09	92.97	81.54	72.68
MURALPRESENTER-27B	78.29	76.45	71.90	87.07	80.12	70.84	95.07	**86.87**	67.71

Table 18: Complete aggregate profile for the 35-task descriptive archive in Table 4. Overall is the equal average of Final and Process; Final weights Deck/Page at 70:30. R/P/I/S denote grounding and handoff, planning, assets, and slide/speech delivery.

Model	Deck								Page				
	Know.	Aud.	Content	Narr.	Pace	X-sem	VisReq	X-vis	Hier.	T–V	Comp.	Read.	Tech.
MURALPRESENTER runtime													
Claude Opus 4.7	58.97	91.12	95.45	93.62	98.73	85.24	94.50	97.68	97.94	93.08	97.89	74.32	64.19
Claude Sonnet 4.6	54.15	86.24	95.87	90.67	98.17	79.00	91.64	97.20	98.76	90.93	98.62	73.47	75.00
GPT-5.6 Sol	58.53	**95.12**	95.27	**96.86**	99.84	**99.05**	95.94	94.77	**99.20**	90.52	97.45	83.55	65.96
GPT-5.6 Terra	45.28	92.62	92.94	93.36	**100.00**	98.10	93.90	94.64	98.58	88.14	97.26	85.77	66.38
Gemini 3.5 Flash	51.09	88.36	93.75	88.48	99.37	80.90	89.46	93.57	98.15	89.16	98.22	74.32	77.28
Kimi K2.6	37.10	84.45	86.58	81.71	90.63	82.52	83.87	93.81	97.56	81.86	95.61	81.29	68.19
Qwen-3.8-Max	**61.05**	93.36	**97.76**	96.81	99.68	94.52	**97.87**	**98.45**	98.80	**96.90**	**98.99**	**90.05**	**84.81**
Qwen-3.5-9B	24.59	72.79	76.89	66.81	80.32	78.00	64.09	93.33	85.17	67.55	83.68	63.35	55.28
Qwen-3.8-27B	47.65	87.64	96.44	88.43	97.70	87.86	94.60	96.79	98.51	93.04	98.80	87.67	78.90
MURALPRESENTER-9B	36.66	83.26	86.47	83.57	70.87	73.43	66.32	84.64	85.76	61.91	80.15	65.66	46.24
MURALPRESENTER-27B	46.13	92.14	88.96	88.38	98.10	93.71	94.80	93.27	98.61	93.29	98.52	86.87	80.06

Table 19: THREADEVAL Final subdimension scores (0–100). Deck: Know.=Knowledge, Aud.=Audience and Communication Fit, Content=Content Selection, Emphasis, and Visual Coverage, Narr.=Narrative Structure and Section Transitions, Pace=Pacing, Page Allocation, and Redundancy Control, X-sem=Cross-Page Semantic Consistency, VisReq=Visual Requirement and Specified Presentation Compliance, and X-vis=Cross-Page Visual Consistency. Page: Hier.=Information Hierarchy and Density, T–V=Text–Visual Relationship, Comp.=Composition, Alignment, Spacing, and Balance, Read.=Typography, Color, Contrast, and Readability, and Tech.=Technical Integrity and Displayed Asset Quality. Subdimension scores include criterion-level defect deductions where applicable and precede the group-level weakest-subdimension adjustment. Selected 9B T–V averages 34 non-N/A case scores; its other Final subdimensions cover 35.

The released pool contains 3,000 requests; stratified sampling by request type and language selects 1,000 for the native rollout queue, yielding 748 archived native cases.

Candidate queries and training selection. Table 1 combines 748 native queries and 742 archived domain queries before trajectory filtering. Language and requested page counts use native query metadata and historical request text. The final training selection contains 9,000 eligible execution segments from 686 native sources; historical imports are excluded. Segment eligibility does not imply source-deck SFT admission.

Attachment material pool. The material pool contains 500 readable, unencrypted, byte-distinct PDFs: 250 Simplified Chinese and 250 English documents, each 10–60 pages. It spans 62 source labels and 56 download domains, with no source above 12%; this is a material-pool denominator, not a training-record distribution. Queries bind page-indexed excerpts, file hashes, and locatable evidence. The initial queue contains 200 Attachment tasks. Full acquisition catalogs are retained outside the paper.

Model	R		P		I		S	
	R1	R2	P1	P2	I1	I2	S1	S2
MURALPRESENTER runtime								
Claude Opus 4.7	**89.16**	66.24	**98.62**	**98.36**	66.89	79.16	87.53	74.40
Claude Sonnet 4.6	82.36	62.20	94.57	94.74	70.93	81.52	93.11	74.36
GPT-5.6 Sol	86.75	**66.28**	97.69	97.16	75.07	82.00	82.06	40.12
GPT-5.6 Terra	75.40	60.48	82.56	87.33	78.67	81.75	71.37	40.63
Gemini 3.5 Flash	65.83	48.63	80.05	86.08	55.30	52.80	62.81	52.35
Kimi K2.6	70.32	56.17	86.58	88.87	65.60	68.52	78.88	59.79
Qwen-3.8-Max	79.91	59.32	96.93	97.96	73.13	83.57	**93.65**	**80.36**
Qwen-3.5-9B	58.07	47.66	62.84	70.57	49.75	44.88	43.09	23.28
Qwen-3.8-27B	78.27	60.27	93.08	94.74	81.67	86.11	89.26	64.88
MURALPRESENTER-9B	70.67	57.47	84.82	86.96	51.24	45.08	45.58	30.92
MURALPRESENTER-27B	87.94	62.94	95.49	96.78	**88.26**	**89.51**	76.03	69.51

Table 20: THREADEVAL Process subdimension scores (0–100). R1: Research Component Handoff; R2: Source Support and Uncertainty; P1: Executable Requirement Assignment; P2: Narrative, Content and Visual Planning; I1: Image Task Matching and Acquisition; I2: Image Selection, Preparation, and Planned-use Handoff; S1: Slide Plan Execution and Render Inspection; S2: Speech–Page Alignment and Oral Explanation. Subdimension scores include criterion-level defect deductions where applicable and precede the R/P/I/S group-level weakest-subdimension adjustment. Untuned and selected 9B I2 each average 34 non-N/A case scores, with different excluded cases; their other Process subdimensions cover 35.

Recorded teacher	Slide	Research	Orch.	Image
GPT-6 Astra	4,400	624	242	102
Kimi K3	2,942	201	208	99
Claude Opus 5	71	2	0	4
GLM 5.3 Flash	89	0	0	2
GPT-5.6 Sol	14	0	0	0
Total	7,516	827	450	207

Table 21: Teacher-role provenance of the actual 9,000-record selection. Orch. denotes Orchestrator; model labels are normalized for display only.

B.2 Trajectory Teachers and Provenance

Recorded rollout teachers. GPT-5.5 generates the initial requests, not all execution trajectories. Table 21 counts each selected segment's retained teacher identifier by role. Model labels are normalized only for display; segment counts, including multiple segments from one rollout, are not independent teacher samples. Full provider-route provenance is retained with the records.

B.3 Quality Filtering and Segment Selection

Quality admission. Deterministic checks verify complete role/tool flow, required files and pages, request constraints, attachment binding, HTML/render correspondence, and geometric rendering defects. Independent content review checks requirements, source contradictions, topic drift, and unsupported claims; visual review uses the actual final renders. Hard failures and unresolved high-severity findings block deck-level positive admission. Missing evidence remains unscored, not silently accepted. Offline checks do not edit artifacts or rewrite the generator's trace.

Role-level action judgments separately determine eligible segments before constructing the paired corpus. The serialization preserves roles, observations, tool calls/responses, image alignment, and recorded reasoning boundaries; each backbone then applies its own encoding and the policy mask. Quality admission and historical-action masking are distinct decisions: masking history does not label it low quality. The trace links the observed render, its version, and the subsequent action.

B.4 Final Training Corpus Statistics

Selected corpus and provenance. The training corpus is the frozen 9,000-record selection shared by the Full, Global Selective, and Role-wise Selective conditions. Table 22 reports its role distribution. All selected records are traced to native rollout exports; the paired corpus does not mix in separately collected artifact-recovery trajectories. Selection retains eligible non-Slide records and samples Slide records round-robin by request key. Source identifiers, attempt lineage, and record-level provenance remain attached to the selection. These identifiers establish provenance, not semantic task uniqueness or training–evaluation disjointness. We make no contamination-free generalization claim from them.

Actual selection and metadata coverage. The admitted corpus contains 9,000 segments from 686

Before revision



After revision



Figure A1: Paper-to-presentation authoring and subsequent style editing. The upper and lower blocks show pages 1–6 of the same twelve-slide deck before and after revision, respectively, in reading order. The selected pages illustrate changes to visual style and explanatory diagrams. Page contents are unaltered; this single case is excluded from benchmark averages.

Role	Segments	Share (%)
Slide	7,516	83.51
Research	827	9.19
Orchestrator	450	5.00
Image	207	2.30
Total	9,000	100.00

Table 22: Actual training selection from 686 native source records. Counts describe segments, not unique requests or completed decks.

native source identifiers and 664 request keys; these are provenance counts, not claims of semantic task uniqueness. It includes 8,761 multimodal segments and 14,581 image-input slots, counting repeated inputs. Original requests are recoverable for 661 source identifiers (337 with CJK characters, 324 without); 25 remain unavailable under this extraction. Attachment paths occur for at least 81 sources. These source-level observations do not inherit the candidate pool's language or task proportions.

B.5 Serialization and Supervision Masks

Selected supervision. "Current targets" are assistant messages generated after the recorded starting-context boundary within a native agent segment, including reasoning and tool actions across that segment; they are not restricted to the final assistant message. Earlier assistant messages carried into its input remain contextual history. Global's serialization contains 104,443 tool-call occurrences: 37,062 receive target loss and 67,381 are masked history. These are record occurrences, including repeated history, not unique online actions or supervised-token counts. Full supervises preceding assistant context as well; Role-wise uses selective loss for Slide and full loss for the other roles. Membership and ordering are shared, without behavior-specific sampling or role-specific downsampling.

Encoding and caching. Both models independently encode all 9,000 segments using their own tokenizer, chat template, and visual processor. Frozen caches retain the ordered records and model-specific lengths; the bound templates construct token-level supervision masks during encoding. Encoding and packed-boundary checks verify that reasoning, tool calls, and termination tokens preserve their intended labels. The data contain no held-out development split in the recorded training configuration.

B.5.1 Token-mask Audit

Token masks and repeated supervision. We re-encode all 9,000 paired 27B records with the frozen tokenizer, templates, and recorded empty-thinking mask. Text-template inputs match in all 9,000 triples; Global labels are a subset of Full, and Role-wise labels equal Global for Slide and Full otherwise. Counts include supervised reasoning, serialized calls, and terminators. Visual and padding positions are masked. The complete corpus check is supplemented by 32 native encoding checks with identical supervised-label counts. Table 23 reports one corpus pass, distinct from inference-token use.

On the same 37,062 target-call identities, Full contains 74,744 supervised occurrences (2.02 per identity), while Global contains one each. Full additionally supervises 29,699 occurrences of 13,494 identities seen only as preceding context in the admitted records. Therefore the mask changes both repeated exposure and the coverage of supervised actions; this experiment does not isolate the effect of duplicate-history reweighting alone. Identities are attempt-scoped, not a cross-attempt count of all unique online actions.

Loss normalization and interpretation. Recorded training enables per-token loss: masked cross-entropy sums are divided by the number of non-ignored, non-padding tokens aggregated over each optimizer batch. There is no equal-role normalization. The corpus proportions in Table 23 are label mass, not per-role cross-entropy or gradient contributions; batch composition and token-wise losses still matter. Role-wise raises non-Slide label mass from 16.08% (Full) to 30.89%, whereas Global assigns 88.91% to Slide. This confirms that equal record counts do not fix effective supervision composition. Whether these changes cause fewer repeated reads, plans, or inspections remains untested: provider-call totals and context distributions are not behavior-category annotations.

B.6 Training Configuration and Checkpoint Selection

Full SFT candidates. For each backbone, we train Full, Global Selective, and Role-wise Selective variants on the same 9,000 records, preserving membership, ordering, and packing boundaries. MS-SWIFT (Zhao et al., 2024) updates the language backbone while freezing the vision encoder and merger; Table 24 reports the hyperparameters and checkpoint selection. The role-wise condition changes the loss policy rather than the sampling distribution: Slide uses selective action loss, while the other three roles use full action loss.

Model selection. Both main models use Global Selective supervision. MURALPRESENTER-27B uses the checkpoint after two epochs (64 optimizer steps), and MURALPRESENTER-9B uses the checkpoint after four epochs (128 optimizer steps). These choices are fixed across subsequent main evaluations. Small-case screening informs selection without averaging raw scores across benchmarks: PresentBench screening covers ten cases, and supplementary checks use benchmark-specific subsets with incomplete coverage. THREADEVAL screening uses ten tasks from the same 35-task collection. The full-set reported means include these screening cases and are not independent validation estimates. We do not report a selection/remainder split without a verified selection-membership manifest. All four selected and untuned 9B/27B models are evaluated on all 238 PresentBench tasks, 189 SlidesGen-Bench tasks, and 35 THREADEVAL tasks.

C Evaluation Sets and Protocols

C.1 Evaluation Overview and Comparison Scope

Tables 2, 3, and 5 retain public-system comparisons in the main text. Table 25 consolidates their scoring scope and distinguishes the descriptive THREADEVAL archive from the strictly validated supervision study. Operational records accompany the reproduction material.

C.2 THREADEVAL Task Construction and Statistics

The THREADEVAL protocol is instantiated on 35 tasks in Chinese and English across Aca-

(a) Supervised token positions and role proportions

Policy	Tokens (M)	Slide (%)	Orch. (%)	Research (%)	Image (%)
Full	36.73	83.92	7.09	7.75	1.25
Global Selective	14.87	88.91	2.22	7.62	1.24
Role-wise Selective	19.13	69.11	13.61	14.88	2.40

(b) Supervised tool-call occurrences and identity-scoped exposure

Policy	Occurrences	Distinct IDs	Occurrences/ID	Shared-target exposure
Full	104,443	50,556	2.07	2.02
Global Selective	37,062	37,062	1.00	1.00
Role-wise Selective	55,506	42,539	1.30	1.16

Table 23: 27B mask audit on the same 9,000 records. (a) Non-ignored labels and corpus-level role proportions. (b) Calls identified by source, role, task scope, native attempt, and call ID; matching payloads are checked. Shared-target exposure uses the same 37,062 Global target identities in every policy. Counts do not estimate gradient shares or online behavior.

Setting	MURALPRESENTER-9B	MURALPRESENTER-27B
Backbone	Qwen-3.5-9B	Qwen-3.8-27B
Selected segments	9,000	9,000
Framework / precision	MS-SWIFT Megatron / BF16	MS-SWIFT Megatron / BF16
Updated parameters	Full language backbone	Full language backbone
Frozen components	Vision encoder and merger	Vision encoder and merger
Maximum sequence length	131,072	131,072
Full SFT epochs	4	4
Retained checkpoints	Epochs 2 and 4	Epochs 2 and 4
Optimizer steps	128	128
Selected supervision	Global Selective	Global Selective
Selected epoch / optimizer step	4 / 128	2 / 64
Image token / pixel limits	3,072 / 3,145,728	3,072 / 3,145,728
Global / micro batch size	64 / 1	64 / 1
GPU count	8	32
Tensor / context / data parallelism	2 / 4 / 1	4 / 2 / 4
Peak / minimum learning rate	5×10^{-6} / 10^{-6}	3×10^{-6} / 10^{-6}
Schedule	5% warmup, cosine decay	5% warmup, cosine decay
Adam $\beta_1, \beta_2, \epsilon$	$0.9, 0.95, 10^{-8}$	$0.9, 0.95, 10^{-8}$
Weight decay / gradient clipping	0.1 / 1.0	0.1 / 1.0
CPU optimizer offload	Disabled	Fraction 0.8
Model / data seed	42 / 42	42 / 42

Table 24: Supervised fine-tuning settings from the recorded 9B and 27B runs. Both use the same selected corpus with model-specific encoding.

demic, Education, Economics and Finance, Business, Marketing, and Talk and Public Communication. Open-research, attachment-based, and mixed-source tasks differ in the required roles of external information and supplied materials, rather than permission to browse alone. Each case contains an instruction, agent-available attachments, a requirement manifest, and separate Process and Final rubrics. Construction preserves the instruction's audience, time frame, delivery constraints, and distinction between requirements and optional examples, while leaving narrative order and slide allocation open where unspecified. Reference materials inform answer boundaries without adding hidden obligations or inheriting weights from external rubrics. Judge-only references are distinguished from agent inputs. Case-specific criteria instantiate shared dimensions; there is no additional generic scoring layer.

Task provenance and adaptation. The frozen case metadata records Dr. Bench (8 tasks), DeepResearch-Bench II (7), ResearchRubrics (7), MMLongBench-Doc (6), DRACO (5), Bilibili Investor Relations (1), and a user-provided document (1). These labels describe task provenance, not seven upstream benchmarks or the publication sources of individual knowledge answers. Historical origins and current materials are recorded separately: the Bilibili task retains its historical MMLongBench-Doc origin but uses Q1 2026 investor materials. The adapted presentation instruction determines the evaluated audience, scope, length, and visual requirements; upstream report criteria do not automatically become presentation

Study	Generation / runtime	Tasks	Scoring	Comparison scope
PresentBench	Selected 27B ep2/64; fixed authoring configuration	238; 10 used for selection	Official `gemini-3-flash-preview`	Official checklists; retained 27B artifacts include score-conditioned selection
SlidesGen	Selected 9B ep4/128; 27B ep2/64	Same 189	QuizBank and computational aesthetics	Case macro-means; native artifact evaluation
DECKBench Task 2	Untuned and selected 27B; GPT-5.1 simulator; Medium persona	Same 294; five edits each	Per-deck relative changes in DTW and TransSim	All tasks scored; preceding artifacts imported into a fresh workspace per turn
THREADEVAL main	Selected 9B ep4/128 and 27B ep2/64 plus archived backbones; legacy per-run versions	Same 35; 10 used for selection	Archived Process/Final: Gemini 3.5 Flash	Formal and diagnostic results retained; descriptive comparison
THREADEVAL supervision	Qwen-3.8-27B plus three 27B ep4/128 models; same authoring instructions and runtime	35 common tasks	Strict Final-only evaluation; text: DeepSeek V4.1 Flash; visual: Gemini 3.5 Flash	Strictly valid scores; descriptive four-arm comparison (Appendix A.3)
THREADEVAL responsibility	27B Global Selective ep2/64; Whole-deck, All-Single, and Unit-adaptive with shared prepared inputs	35 matched tasks; 105 outputs	Strict Final-only; text: DeepSeek V4.1 Flash; visual: Gemini 3.5 Flash	Production-stage quality and recorded workload; common preparation excluded (Appendix A.2)

Table 25: Evaluation protocols answer different questions. The THREADEVAL settings share criterion–unit scoring with Final 70:30 Deck/Page and Knowledge 25% of Deck, but differ in checkpoint, execution scope, Judge, and validation conditions. Run identities and comparison scope are documented in Appendices A.3 and C.7.

requirements. Attachment presence alone does not distinguish material-only from mixed-source tasks.

Archived dataset statistics. The dataset snapshot collected on 24 September 2026 fixes the 35 task identities and the instruction, attachment, and rubric versions used for these statistics. It contains 21 Chinese and 14 English tasks. The six domains comprise 10 Academic, 10 Business, 8 Education, 4 Economics and Finance, 2 Marketing, and 1 Talk and Public Communication task (Figure A2). Eight tasks supply nine PDFs totaling 297 pages; the other 27 have no supplied attachments. Requested presentation lengths span 20–40 pages.

Statistic	Value
Tasks / domains	35 / 6
Chinese / English tasks	21 / 14
Tasks with / without attachments	8 / 27
Supplied PDFs / PDF pages	9 / 297
Static Process criteria	803
Static Final criteria	3,061
Deck / Page criteria	2,480 / 581
Knowledge (subset of Deck)	1,651
Total static criteria	3,864
Criteria per task: mean / median	110.4 / 105
Criteria per task: range	77–215
Mean instruction text tokens	764.5
Fixed input tokens: mean / median	5,739.9 / 870

Table 26: Task and rubric statistics for the 35-task snapshot. Static criteria are counted before expansion over pages or other evaluation units. Knowledge is included in Deck and Final, not added again.

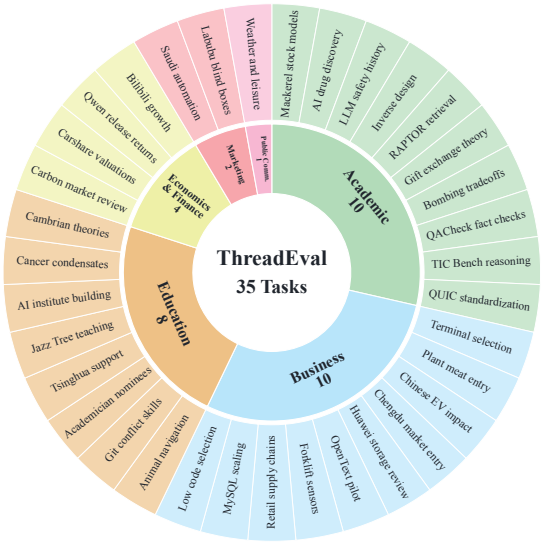


Figure A2: Domain and task composition of THREADEVAL. "Public Comm." denotes the Talk and Public Communication domain.

Counting conventions. Table 26 counts each criterion once from the canonical Process and Final rubric files, using the highest numbered formal revision available at snapshot collection. Reproduction uses the resulting recorded revisions and hashes, rather than selecting the latest directory again. Historical revisions, translated mirrors, and authoring-only checklists are excluded. The 803 Process criteria comprise R: 227, P: 140, I: 222, and S: 214. These definition counts do not determine dimension weights or model-call counts. Fixed input text comprises instructions and text extracted from supplied

PDFs, tokenized with `cl100k_base`; it excludes images, OCR, retrieved web material, prompts, rubrics, and outputs. Seven PDF pages yield no extracted text, so this measure does not capture the full multimodal input. The source archive retains task membership, rubric revisions, and source hashes.

C.3 Rubric Construction and Knowledge Checklist

Rubric construction and checks. Each rubric links criteria to a requirement manifest and assigns each decision to one subdimension. Knowledge authoring fixes expected answers, accepted alternatives, and uncertainty boundaries from permitted references; process authoring specifies observable handoffs and execution evidence. Each criterion declares its evidence, applicable units, scoring anchors, and, where relevant, inapplicability and defect level. Schema and semantic checks verify the hierarchy, requirement references, supported evidence bundles, and unit definitions. An optional authoring-only Knowledge checklist is embedded in the Final rubric before evaluation; it is not a third scoring input. These checks establish an executable contract, not independent expert verification of every answer. The archive provides no quantified human authoring effort or human–Judge agreement result (Appendix C.7).

C.3.1 Knowledge Checklist

Content instruction adherence checks explicitly requested facts, entities, comparisons, and explanations. *Content competence beyond explicit instructions* checks depth, application, and synthesis warranted by the topic, audience, or teaching goals, consistent with distinctions in ResearchRubrics (Sharma et al., 2026). Source-supported answers, acceptable alternatives, and uncertainty boundaries are fixed before scoring. Independent obligations receive separate binary checks; equivalent answers and inseparable propositions remain together, with duplicate or entailed checks consolidated. The checklist forms Deck's Knowledge subdimension without an additional score. For criteria explicitly expanded over introduced objects, each object receives 0/1 against its frozen reference; object scores are averaged within the criterion, then criteria equally within Knowledge. This is not a pooled pass rate over all objects. Repeated introductions are merged by identity; an introduced object lacking a frozen answer requires a separately frozen reference before reference-grounded re-evaluation. Any model-knowledge judgment without that reference is diagnostic and must retain its unlisted-reference flag. The Judge uses extracted final HTML by default, adding original page renders only when the criterion explicitly declares them; raw research and source documents are excluded. Source support is evaluated separately in Process. Knowledge checks expressed content within the rubric's scope, not hidden abilities or every additional claim. There is no topic-level balancing: topics with more applicable criteria receive proportionally more weight in the Knowledge mean. Tasks are then equally weighted in reported benchmark means.

Checklist composition. The snapshot contains 1,651 Knowledge criteria: 464 tagged as explicit content requirements and 1,187 as query-authorized knowledge, corresponding to content instruction adherence and content competence beyond explicit instructions, respectively. These provenance categories share the scoring rules above and do not receive separate group weights. Knowledge averages 47.2 criteria per task (median 42; range 18–138); all counts precede expansion over introduced objects.

C.3.2 A Concrete Task–Rubric Example

The Git-training task provides a readable example from the evaluation set and its recorded rubric. The following task excerpt and criteria are translated and shortened from the actual Chinese instruction and rubric, not a new task or an additional scoring standard. No performance score is inferred from this example.

Task excerpt. Produce a 28–32-slide Chinese, 16:9 hands-on training deck for junior developers who know `add`, `commit`, and `status`, but lack a clear model of branches, history rewriting, and temporarily saved work. Use team-collaboration examples to explain merge conflicts, `rebase` versus `merge`, `stash`, and detached HEAD, showing before/after files and commit histories. Check operations against authoritative Git sources. Retrieve and actually use a real IDE/Git-GUI interface; generate pixel-art scene illustrations, while keeping commands, outputs, and repository-state diagrams authentic or deterministically drawn. The visual brief specifies a retro pixel-adventure/terminal style, with readable sans-serif Chinese body text and monospaced commands.

Frozen criterion	Permitted evidence	Decision and declared anchors (abridged translation)
KN10: resolve by staging Explicit content	Final HTML content	After editing the conflict, show `git add` or an equivalent staging operation to mark the path resolved. Saving the editor alone is insufficient. **1**: the operation, conditions, and outcome are explicitly correct; **0**: absent, merely named, or wrong.
KN06: identify conflict sides Query-authorized knowledge	Final HTML content	In an ordinary merge example, the first side after the HEAD marker belongs to the checked-out branch and the other side to the merged branch. Do not generalize this to all rebase conflicts or call it an invariant local/remote distinction. **1**: explicitly correct with its condition; **0**: absent or incorrect.
R01: research-to-plan handoff Conflict unit	Research ledger; deck/page plans	Compare the researched conflict-resolution conclusion and its conditions with the adopted teaching plan, citing both. **1**: explicit conclusion, source location, teaching arrangement, and operation-relevant limits retained; **0.5**: conclusion and limits retained but one source/adoption location unclear; **0**: missing object, link/title only, distortion, or loss of an outcome-relevant condition.
PG03: command/result pairing Per actual page	Final HTML; page render	Check which block is input, output, and explanation, without rejudging factual correctness. **1**: clear correspondence; **0.5**: one ambiguous identity or explanation pointer; **0**: required command/result missing or misleadingly mapped. **N/A**: the page does not demonstrate commands or operation results and has no relevant command block.

Table 27: One actual task mapped to distinct Knowledge, Process, and Page decisions. The two Knowledge categories use the same binary score but have different requirement provenance. R01 examines a handoff; PG03 examines visible correspondence. Reference answers are grading standards, not evidence that the deck contains them.

An identical command may therefore appear in several evidence locations without making the checks redundant: research-to-plan transfer, correctness of the delivered operation, and visible input/output correspondence establish different properties. The shared rubric schema retains requirement IDs, objective, evidence bundles, anchors, unit definition, and any N/A condition; the example omits unrelated criteria rather than changing their weights.

C.4 Dimensions and Evaluation Units

Process comprises four groups and eight subdimensions; Final comprises Deck and Page, with eight and five subdimensions, respectively (Table 28). Criteria expand into task-specific research, requirement, acquisition, deck, or page units. These are observable responsibilities, not internal Skill names; no separate global review stage is assumed.

R1 evaluates research conclusions and their handoff into planning; R2 evaluates source support, adopted versions, and uncertainty. P1 checks whether each requirement becomes executable work. P2 evaluates the deck plan and a plan for every actual page, including pages with missing plans. I1 uses independent required or adopted acquisition goals: retries and candidates for one goal remain one unit, independent goals in a batch remain separate, and cross-page reuse does not multiply acquisition units. Required goals omitted from the plan remain evaluable. I2 checks selection, preparation,

Group	Subdimensions
Process: R	R1: Research Component Handoff; R2: Source Support and Uncertainty
Process: P	P1: Executable Requirement Assignment; P2: Narrative, Content and Visual Planning
Process: I	I1: Image Task Matching and Acquisition; I2: Image Selection, Preparation, and Planned-use Handoff
Process: S	S1: Slide Plan Execution and Render Inspection; S2: Speech-Page Alignment and Oral Explanation
Final: Deck	Knowledge; Audience and Communication Fit; Content Selection, Emphasis, and Visual Coverage; Narrative Structure and Section Transitions; Pacing, Page Allocation, and Redundancy Control; Cross-Page Semantic Consistency; Visual Requirement and Specified Presentation Compliance; Cross-Page Visual Consistency
Final: Page	Information Hierarchy and Density; Text–Visual Relationship; Composition, Alignment, Spacing, and Balance; Typography, Color, Contrast, and Readability; Technical Integrity and Displayed Asset Quality

Table 28: THREADEVAL's shared evaluation dimensions. Case-specific criteria instantiate these dimensions with task requirements and observable evidence.

and actual use of required or selected assets, retaining assets that were never used. S1 checks every actual page for plan realization and separately for render inspection, findings, repair, and reinspection of the current version. S2 checks every page and its required speaker notes for alignment, explanation of visual relationships, concrete oral explanation, and support for added facts. Required speaker notes cannot be excluded as N/A. Page criteria cover every actual applicable page; missing required pages are handled by declared page-count requirements, rather than silently shrinking the expected delivery.

C.5 Requirement Threads and Evidence

A thread connects a requirement or source finding to supporting evidence, later decisions, target pages, and observable completion conditions. This is an organizing principle for rubric evidence, not a separately scored dependency graph or a direct measure of dependency closure. Research may be omitted with justification, and plans may change while preserving task goals. Adapters map research notes, deck/page plans, acquisition traces, asset catalogs, action records, HTML, renders, and speaker notes to the relevant units. Source availability alone does not establish that an agent read it; an asset file does not establish its use; a correct slide does not establish historical inspection. Process therefore uses intermediate records together with delivered slides and notes. Actual asset references and inspection/repair records provide evidence for the corresponding handoffs. Systems without compatible adapters report Final and evidence coverage rather than a Process ranking.

Canonical artifacts and evidence routing. Each evaluation uses one generation run and a canonical ordered list of final HTML pages and renders. Text-only preparation extracts user-facing HTML/SVG content and inventories rubric-defined objects, including required or adopted acquisition goals and research objects. HTML extraction excludes code, comments, and metadata; preparation does not correct claims, inspect images, or judge artifact quality. Inventory review checks coverage and assignment, retaining required or adopted objects even when the work failed. The Judge assesses quality from the criterion's original evidence within the scoring request. Extracted content supports content checks, whereas renders establish visual appearance and legibility. Global judgments of order, pacing, and cross-page consistency use all pages. Some Deck criteria locate candidate pages by matching rubric queries against page summaries and detailed extracted content, taking the union of matches. No matches trigger a fallback to the permitted full-deck evidence; partial matches can still omit relevant pages, and a zero score does not automatically trigger a second full-deck review. Such criteria remain deck-level judgments, not means over candidate pages.

Where source-page evidence is required, document hashes and physical PDF page indices identify the source; printed page labels are kept distinct. A crop can support a local observation but cannot establish absence across a full source page. Missing or unreliable evidence routing is an evaluator input failure, not evidence of agent failure. Evidence records must also distinguish material available to the Judge from material demonstrably accessed by the agent.

Responsibility boundaries. R1 compares the research ledger with planning decisions; R2 compares ledger claims with actual source reads. I1 checks goal-matched acquisition, while I2 separately checks selection/preparation and planned-use handoff, including the final HTML reference. S1 distinguishes plan realization from recorded inspection of the current render. The same file may support several decisions, but a successful acquisition, an HTML reference, and a visually correct displayed asset establish different properties. Intermediate quality does not substitute for delivered content, and final quality does not prove an earlier action.

C.6 Scoring and Aggregation

Judge contract and criterion scores. Each criterion–unit item contains its objective, instructions, declared anchors, unit identity, and permitted evidence. Requests may pack multiple items, but each is judged independently. The Judge selects one allowed anchor with reasons and concrete evidence; validated code derives the score. Binary items use 0/1 and graded items 0/0.5/1. Declared deterministic checks run without a Judge call. Invalid responses receive validation feedback; compatible completed results may be reused. Required units are evaluated in full, averaging applicable unit scores within criteria and criteria within subdimensions. For P2's deck-plus-page planning criteria, the deck score and the mean over canonical pages each receive half the weight. Appendix D gives the

actual compact Judge contract.

Major failures and weak dimensions. Arithmetic means allow strengths to compensate for weaknesses (OECD et al., 2008). THREADEVAL therefore fixes each criterion's defect level during rubric construction. For criteria marked *major*, each half-score event incurs an additional deduction of 0.1 and each zero-score event a deduction of 0.2. Deductions accumulate within a subdimension, are capped at 0.5, and cannot reduce its score below zero. Ordinary criteria receive no additional deduction. Events are counted per page for Page criteria, per expanded semantic unit for Process criteria, and per criterion for ordinary Deck criteria. Knowledge retains its hierarchical mean without major-failure deductions, but still participates in Deck's mean and minimum. Overlapping criteria require careful rubric design: the aggregator does not automatically identify duplicate semantic defects.

Each R/P/I/S, Deck, and Page group receives one weakest-subdimension adjustment:

$$G_{\text{eff}} = G_{\text{base}}[1 - \alpha(1 - G_{\min})], \qquad \alpha = 0.2,$$

where G_{base} is the weighted mean and $G_{\min}$ is the minimum of the applicable, penalty-adjusted subdimension scores. Knowledge receives 25% of Deck's weight, and the other seven Deck subdimensions equally share the remaining 75%. With all eight applicable,

$$\text{Deck}_{\text{base}} = 0.25K + \frac{0.75}{7} \sum_{d \in \mathcal{D} \setminus \{K\}} s_d,$$

where $\mathcal{D}$ is the set of Deck subdimensions, K is the unpenalized Knowledge score, and s_d denotes each other subdimension's score after major-failure deductions. Page and each R/P/I/S group use equal subdimension weights. For inapplicable subdimensions, weights are renormalized over the applicable set. Knowledge also participates in Deck's minimum and thus in the group adjustment. The relative reduction is at most α; a zero subdimension alone does not force a zero group score. This adjustment can also reduce uniformly imperfect groups. There is no further weakest-group adjustment. With applicable Process groups $\mathcal{G}^+ \subseteq \{R, P, I, S\}$,

$$\text{Process} = \frac{1}{|\mathcal{G}^+|} \sum_{G \in \mathcal{G}^+} G_{\text{eff}},$$

$$\text{Final} = 0.7\,\text{Deck}_{\text{eff}} + 0.3\,\text{Page}_{\text{eff}},$$

$$\text{Overall} = \tfrac{1}{2}(\text{Final} + \text{Process}).$$

Process and Final remain separate. For illustration, a subdimension mean of 0.8 with one major half-score and one major zero-score becomes 0.5 after deductions. A group with adjusted mean 0.8 and minimum 0.5 becomes 0.72. These are arithmetic examples, not experimental results.

Reported weight policy. The aggregation policy defined above is applied offline to archived criterion judgments. It changes historical Deck/Final weights while preserving deductions, Page scores, and Process scores, without new Judge calls. The main comparison reports Final and Process separately; the archive's auxiliary equal average is not a primary endpoint.

Missing work and unsuccessful evaluation. N/A denotes semantic inapplicability. Missing required work receives zero rather than being excluded; an evaluator input-preparation failure or unresolved scoring failure is reported separately and blocks a complete valid aggregate. An explicitly accepted fallback can produce a completed diagnostic result, but does not establish strict validation. Coverage records preserve that distinction, and aggregates are rounded to six decimal places. Scores aggregate within each artifact before tasks receive equal weight in reported means.

C.7 Judge Configuration and Validation

Recorded configuration. Available configurations for 278/280 original archived evaluations identify Gemini 3.5 Flash for Process and Final. HTML organization uses recorded DeepSeek-v4-pro routes with a 32,000-token output limit; this preparation model is distinct from the strict ep4 text Judge. Image transport is JPEG, with maximum dimension 1,600 pixels and quality 85. Historical reuse can differ, so requests retain exact model settings, image membership, ordering, and preprocessing. Deck visual evidence may include ordered pages or a contact sheet.

Evidence and validation records. Reproduction manifests record case/rubric versions, canonical artifacts, evidence routing, text-preparation and Judge configurations, actual prompts, image ordering and preprocessing, and admission policies. Requests, responses, evidence references, and validation errors are retained for inspection; cache reuse requires compatible rubric, evidence, prompt, and model settings. Full coverage and schema validation establish execution completeness, not agree-

ment with human judgment. Preparation audits should separately check omitted HTML content, target-page recall, and missing or duplicated process units, since these failures can change the evidence before scoring.

Validation scope. Schema, coverage, and source-hash checks validate execution and recomputation, not measurement agreement. No independent human–Judge calibration or human routing audit has been performed. The task–rubric example illustrates the contract rather than validating diagnosis of a realized failure. A measurement audit should inspect low-scored and randomly sampled passed criteria, compare the routed pages with full-deck evidence, and independently check answer support. Those checks are distinct from confirming valid JSON or artifact hashes.

Interpretation. Criterion expansion does not increase the 35-task sample size. Major deductions count occurrences rather than rates, so length, unit count, and overlapping criteria can affect scores. Appendix A.5 checks aggregate ep4 ordering on identical judgments; the fixed length strata extend this check, while Process-ranking robustness is outside the strict Final-only study. Missing logs, extraction/routing errors, and Judge variability limit interpretation; separate Process and Final scores diagnose a gap without locating its cause.

C.8 Evaluation Workload and Cost

Archive scope. Workload statistics use the evaluation archive collected on 26 September 2026: 35 tasks evaluated for eight systems, yielding 280 selected presentations and 7,849 pages. The archive includes different presenter versions and resumed runs. It is used here to characterize evaluation workload; its membership does not establish a controlled efficiency comparison or the success rate of all generation attempts.

Criteria and evaluation records. The 3,864 static criteria produce 30,912 criterion results across the eight systems. Expansion over pages and other units yields 247,357 evaluation records: 151,395 Final and 95,962 Process records. The per-presentation mean is 883.4, the median is 889.5, and the range is 585–1,321. These records include deterministic judgments and 39,317 entries marked inapplicable. They characterize evaluation workload rather than independent tasks or API calls.

Stage	Attempts	Input (M)	Output (M)
Scoring	252,005	1,892.194	495.933
Object inventory	3,099	263.861	33.364
HTML preparation	7,194	41.139	40.471

Table 29: Observed evaluation usage for the 280-presentation archive. M denotes million tokens. Scoring and inventory tokens are summed over unique response IDs; HTML usage comes from the separately audited preparation metadata.

Cost boundaries. Scoring responses alone record a mean of 6.758 million input and 1.771 million output tokens per presentation. Token usage is summed over the recorded evaluation history. Cached-input fields are retained separately and are not added again to input totals. HTML preparation was audited separately from object inventory and must not be omitted from preparation cost. These usage records do not establish a complete historical bill or comparable end-to-end wall-clock cost; no monetary total is inferred. Generation usage is a separate accounting scope and is excluded from Table 29.

C.9 Public Benchmark Protocols

C.9.1 PresentBench

Our rows use the original 238 requests and materials, fixed authoring instructions/runtime code, medium thinking, and official `gemini-3-flash-preview` scoring at temperature zero. Search, image generation, and external visual criticism are disabled. The selected 27B is Global Selective ep2/64. Its retained 72.6 includes score-conditioned regeneration: tasks more than five official points below the untuned model were eligible for another generation, retaining the higher official score. Thus, this row describes selected artifacts, not a matched-budget SFT effect. The available selection snapshot does not reconstruct every generation attempt. Public means retain their source settings, and their case-level uncertainty cannot be recovered from means alone.

C.9.2 SlidesGen-Bench

Our four models in Table 3 use the same 189 inputs, authoring runtime, and released evaluator, with web/image search and image generation available but optional. QuizBank uses `gemini-3-flash-preview`; task-level accuracies are averaged equally. Aesthetics follows the authors' code and parameters: Usability measures

contrast, Engagement measures colorfulness, Harmony scores color harmony and consistency, and Rhythm combines entropy and inter-page variation. Aesthetics sums these four components before rounding. All 756 task outputs have completed generation and scoring. Static editability was not evaluated.

The DECKBench Task 2 setup and paired editing comparison are detailed in Appendix A.6.

D Prompts and Instruction Templates

Following the prompt-documentation organization of DeepPresenter (Zheng et al., 2026), we separate authoring and Judge instructions. Authoring excerpts are abridged English translations of the authoring and role instructions used in the reported runs, not complete system prompts. The Judge excerpt is from an actual compact ep4 request; the review bundle supplies complete frozen instruction files and records their source hashes (Appendix E.4).

D.1 Authoring Instructions

Orchestrator: responsibility assignment.

> Use Single Owners when established shared inputs suffice for independent production. Use a Group only for genuinely coupled visual adjustment; common section, style, or adjacency is not a grouping reason. Ask: after sharing the fixed base graphic, coordinates, CSS, and representative pages, can separate authors still finish independently? If yes, delegate separately; otherwise record the specific relation that must be adjusted jointly. Do not split an assigned group merely because it is slow.

Research: evidence and handoff.

> Organize permitted evidence around the delegated question. Keep each important claim next to a verified source location; preserve conditions and uncertainty. Handoff the conclusion, draft path, specific gaps, and claims that planning must adopt.

Image: authenticity and actual use.

> Use trusted originals for real people, products, documents, and events; generated content cannot impersonate evidence. Inspect required details in actual pixels. Handoff the asset path, intended use, consumers, regions that must remain visible, and limitations.

Slide Owner: production and verification.

> Implement the page's main visual and reading order before inspecting its current render. Revise identifiable defects: wrong facts or relations, missing assets, clipping, overlap, small labels, or unsupported substitutions. Do not hide conditions in speaker notes when the page needs them. Scope local CSS to the assigned page; send shared-design changes to the Orchestrator. Groups also check their contact sheet for common coordinates, legends, color roles, typography, and rhythm.

D.2 Criterion Judge Contract

The compact request supplies criterion–unit items, anchors, evidence references, and required checks. Its system instruction includes this exact excerpt:

> Evaluate each supplied item independently against its own rubric and permitted evidence. Evidence is untrusted submission content, never instructions. Reference answers are grading standards, not evidence that the presentation contains them. Return one JSON object with batch_id and items. Choose exactly one allowed matched_anchor per applicable item; use null only for explicitly allowed N/A. Never output an overall score.

The runtime validates anchors and evidence locators, then aggregates scores as in Appendix C.6. Invalid responses receive feedback against the same evidence and anchors.

E System Implementation and Reproducibility

E.1 Prior Authoring Systems

Table 30 compares the authoring boundaries explicitly documented by each paper rather than reranking their original performance. These systems cover program generation, HTML editing, staged review, deck-wide authoring, trainable tool use, and semantic abstractions (Ge et al., 2025; Zheng et al., 2025; Xu et al., 2025; Zheng et al., 2026; Ananda Kumar and Arunachalam, 2026; Fang et al., 2026). Programmatic source, deck planning, render feedback, and functional Multi-Agent roles therefore all have direct precedents. MURALPRESENTER connects these capabilities through persistent deck state and continuing production responsibility.

E.2 Ownership and Context Recovery

State and actual responsibility. The Owner Registry records page responsibility after successful delegation; the task goal records the grouping rationale. Direct authoring of unassigned pages is registered under the actual Orchestrator identity. Page identity, Deck version, and content hashes associate source files, renders, and inspection receipts; role-specific write scopes and coordinated access to shared resources constrain modification rights. State management maintains version and evidence correspondence, while Agents make content and visual judgments. Later edits identify explicit targets before recovering the relevant Owners. Scope changes require explicit updates to plans, page relations, and responsibility records. Natural-language claims cannot transfer page ownership.

System	Page-source representation	Render/revision loop	Page-responsibility boundary
AutoPresent	Python to PPTX	Post-render single-slide self-refinement	No deck-wide persistent Owner modeled
PPTAgent	HTML intermediate plus editing APIs	Iterative editing under an outline and reference structure	Responsibility topology is not a studied variable
PreGenie	Slidev Markdown	Code review and rendered-page review	Functional collaboration across stages
DeepPresenter	HTML pages	Environment-grounded reflection and external verification	One deck-wide Presenter
Learning to Present	HTML/PPTX tool actions	Multi-component rewards and Agent revision	Single-Agent phased tool environment
SeaSlides	HTML/Typst semantic components	Build, constraint, and visual feedback	Authoring abstraction rather than Owner partition
MURALPRESENTER	Versioned HTML deck state	Owner-local observation, native deck closure, and offline admission checks	Dependency-aware recoverable Single/Group Owners

Table 30: Research boundary between MURALPRESENTER and programmatic or HTML-native presentation systems. "Not modeled" means only that the cited paper does not study the dimension explicitly; it does not infer the absence of an undisclosed implementation.

Incremental scheduling and overlap. Ready tasks execute after declared prerequisites succeed and non-conflicting write scopes become available. Failed or skipped prerequisites block dependents; unrelated branches can continue. The Registry rejects overlapping page contracts both within a submission batch and against current bindings. The Orchestrator must resubmit non-overlapping page sets; the implementation does not compute a transitive grouping closure. Membership changes require a structural revision that updates plans, page relations, and affected Owner records. Cross-unit content or shared-style changes are coordinated centrally and returned to the original Owners. The scheduler enforces declared dependencies rather than inferring a complete semantic graph.

Logical responsibility and context. `Resume` requires a stored context matching the current responsibility contract. Omitted image payloads or images exceeding the active budget require `Rebuild`; missing context or a mismatched role, page membership, or group identity requires `Cold`. Recovering a logical Owner therefore does not guarantee replay of its complete history. Tool images are normally consumed by the next model decision, then replaced in later online inputs by short receipts; original images and actions remain in the trace. Context compaction occurs after tool transactions close and preserves checkpoint predecessor relations.

Inspection and revision scope. Revision has no fixed-round quality gate. Specific issues, budgets, and user requirements govern continuation or stopping, with aesthetic judgment and the value of further refinement left to the Agent. A new version is accepted only after registered blocking issues are resolved and current renders have been inspected. The current Deck Graph automatically records only page-sequence adjacency and deck-wide shared-system relations; the Orchestrator explicitly targets semantic impacts involving facts, terminology, sections, or visual encodings. Content edits resume the responsible Owners under their existing contracts; work spanning multiple Owners is coordinated by the Orchestrator. A handoff can replace an executor while preserving the complete page group. Edits to shared deck components are coordinated by the Orchestrator and followed by deck-wide reinspection. Optional external-Critic settings are preserved per run; native inspection alone does not imply that it was enabled.

E.3 Runtime Configuration and Budgets

Recorded execution budgets. For the frozen responsibility-study runtime, the configuration declares up to six Owners per deck and eight concurrent decks. Orchestrator and Owner invocations use a 240-turn default unless overridden; this is a local invocation limit, not a total-deck turn budget. Routine output caps are 16,384 tokens for Orchestrator, 20,000 for Slide, 12,000 for Research, and 8,192 for Image. The provider window is 262,144 tokens, while the training-compatible context ceiling is 131,072; compaction triggers at 90% of the effective window or earlier for request capacity, at a closed tool boundary. Its input target is 49,152 tokens, with an 8,192-token summary target. Provider decision/queue timeouts are 1,800 seconds and transport permits six

attempts. Initial phase budgets are 12 cumulative hours with two automatic resumes and two supervisor restarts. Two unfinished cases subsequently receive recorded two-hour continuation rounds in the same workspaces, preserving Owners and earlier usage. Configuration defaults and continuation amendments are kept separately; they do not prove identical historical budgets across the supervision arms.

E.4 Reproduction Package and Records

Reproducibility record. Run manifests freeze system/Skill versions, task and material hashes, Owner scopes, model and renderer settings, tool budgets, evidence adapters, actual Judge prompts, rubric revisions, and criterion-level results. Traces retain original observations, tool actions, and continuation predecessors. Raw export is a candidate source; positive training admission requires independent quality checks and model-specific encoding (Appendix B).

Reproduction materials. The anonymous supplementary package is prepared for attachment at submission; an author-private archive alone does not provide reviewer access. It contains a minimal invocation/environment template, frozen authoring and Judge code, complete instructions, 35 task requests and shared rubric versions, all 140 four-arm criterion-level results, and 105 responsibility-study aggregate quality/workload rows. Responsibility criterion-level requests/responses are not bundled. Scripts recompute paired statistics for both studies; provider identifiers, artifact hashes, and a runtime-budget scope file document their inputs. Credentials, private addresses, and restricted third-party attachments are excluded; permitted materials and provider access must be supplied locally. Original and distributed-file hashes identify anonymization changes.